

Country Banks and the Panic of 1825

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Abstract

The Panic of 1825 was one of the world's first international financial crises. We document how this crisis spread from London banks to England's real economy. England's correspondent banking network propagated trouble in sovereign debt markets to small banks outside of London and ultimately to non-financial firms. Using exogenous variation in district-level exposure to the crisis, we show that bank failures led to a substantial number of bankruptcies among non-financial firms. We explore two potential mechanisms by which the financial system affected the real economy: a fall in aggregate demand and a credit supply shock.

1 Introduction

The role of finance in England’s industrialization remains the subject of active debate. Scholarly opinion ranges from private finance having little role in spurring the Industrial Revolution (Temin and Voth, 2013) to promoting a growth oriented structural transformation of the economy (Heblich and Trew, 2019). Clarifying this relationship is essential to understanding both the drivers of the Industrial Revolution and the mechanisms through which financial development supports real economic activity.

This paper studies how bank failures during the Panic of 1825 affected local economic outcomes. The isolated provincial economies of England in the 1820s provide an ideal setting to identify the costs of bank failures on local economic activity. Economic historians have debated the role of country banks - banks outside of London - in supporting industrialization and economic growth during the Industrial Revolution but a lack of data and proper identification have both been major barriers to answering this question. To address these challenges, we use newly collected data on the universe of English country banks from 1820-1830 and data on local bankruptcies during this period. We provide the first causal evidence that bank failures during the crisis contributed to the decline in real activity. We present historical evidence for two mechanisms: an aggregate demand channel and a credit supply shock channel. In doing so, we also shed light on the role England’s banking system played during the Industrial Revolution.

The Panic was caused by the first cases of sovereign default in international capital markets (Bordo, 1998; Dawson, 1990; Morgan and Narron, 2015). Like many modern financial crises, the Panic occurred after a large number of new securities appeared in financial markets. Surprising news about the low quality of these assets caused runs on the unregulated financial institutions in London that were exposed to them. Through the correspondent banking network of relationships between London banks and small country banks in English towns outside of London, the crisis exerted significant stress on the country banks. Credit and financial intermediation both within and outside of the financial center (London) contracted. Since the source of the initial shock was external, it provides a natural setting to study the effect of a banking panic on real activity.

The crisis was significant from a macroeconomic perspective: among Britain’s banking crises over the past 200 years, [Turner \(2014\)](#) puts only the Panic of 1825 on par with the Great Recession of 2007-8 in terms of financial distress and output costs. More than 10% of England’s country banks went bankrupt during the Panic and real activity declined dramatically; bankruptcies more than doubled while construction activity, measured by brick production, fell by 30% ([Shannon, 1934](#)); the value of exported cotton manufactures fell 20% while the quantity of raw wool imports fell by more than 50% ([Gayer et al., 1975](#)).

Our data links each bank outside of London (“country banks”) to a particular “correspondent” London bank (“agent”). As part of these relationships, country banks left large deposits with their London agent for investments and clearing purposes. Failure of a London agent therefore had a significant impact on the operations of country banks with accounts with that agent. During the Panic, only agent banks were directly exposed to the sovereign debt crisis, though their failure caused a shock at their client country banks. Our instrumental variable approach compares the number of non-financial firm failures per capita in districts with and without country bank failures, instrumenting for country bank failures using district-level exposure to the sovereign debt crisis through the correspondent banking network. The first stage shows that transmission of financial stress occurred through the banking network. The second stage demonstrates that local bank failures led to the failure of non-financial firms.

We find that economic activity declined in areas with bank failures, suggesting that a bank provided a necessary service to local firms. The Panic resulted in failures of local country banks well beyond the London banks directly exposed to the sovereign debt crisis that instigated the Panic. Our instrumental variable regression results imply a 238% increase in local bankruptcies per capita due to a country bank failure driven by a London agent failure. This result is robust to a range of variations on our main specification. We also observe a financial contagion effect in a second phase of the crisis. Bankless districts were largely insulated from the rise in bankruptcies, providing further proof that the crisis spread through the financial network. We discuss two possible ways financial stress on the country banks was transmitted to local economic conditions: (i) a drop in aggregate demand due to lost household wealth; (ii) a negative credit supply shock, particularly to working capital

lending.

Related Literature This paper contributes to two strands of literature. First, the role of finance, and of country banks in particular, during England’s first industrial revolution is not well understood. Contemporaries pointed to the high failure rate of country banks as evidence that these banks harmed the towns they served and excessive note issuance by country banks was blamed for credit boom and bust cycles in the early 19th century. Some have argued that restrictions on the maximum number of partners and the maximum interest rate banks could charge caused the Industrial Revolution to be “financed out of the pockets of tinkers and manufacturers, not through bank lending” (Calomiris and Haber, 2014). Others argue that public finance of the Napoleonic wars largely crowded out private finance (Murphy, 2014; Temin and Voth, 2013).

Those arguing for the importance of private finance include Crouzet (1972), Mathias (1973), and Pollard (1964), who point out that banks’ provision of short term credit freed up internal profits for reinvestment in longer term capital. Indeed, these plough-backs were the primary source of fixed capital formation during this period, and would not have been possible had firms been obliged to meet their short-term needs with profits.¹ Hebllich and Trew (2019) find that employment in the financial sector in 1817 was associated with structural transformation and industrialization by 1881. This paper complements those findings by focusing on the short term consequences of losing financial services, which we find to be large, suggesting firms relied on the financial services country banks provided. We also find a positive correlation between a district’s access to the financial sector and local patenting.

Second, it adds to the literature measuring the cost of financial crises for non-financial firms. Chodorow-Reich (2014) and Fernando et al. (2012) use a similar identification strategy to study the effects of particular institutions’ collapse on the availability of credit during the Great Recession. Amiti and Weinstein (2018) and Mian and Sufi (2008) use matched firm-lender data in Japan and Pakistan, respectively, to identify firms’ credit supply shocks. Peek and Rosengren (2000) study the effects of the 1990s Japanese banking crisis on construction

¹Brunt (2006) suggests country banks also played a role in industrialization by funding fixed capital investment but evidence on this point is sparse. Policy changes after the Panic (described in detail at the end of section 2.2) make it difficult to study the longer-term causal effects of bank failures on town-level outcomes.

activity in the U.S. using local variation in Japanese bank penetration. A concern in many of these studies is the extent to which firms can switch banks to avoid loan supply shocks. The highly localized nature of English lending and rigid bank-firm relationships during this period provide an ideal setting to isolate the effects of financial stress when switching is not possible.

The focus of much of this literature has been on identifying the effects of a credit supply shock. Yet financial crises that disrupt the payment system or otherwise negatively affect household wealth may also feature reductions in aggregate demand. One study that focuses on the local demand effects of banking crises is [Huber \(2018\)](#), though the demand effects he documents for Germany are essentially second-round effects of the credit supply shock due to employment losses. We instead highlight the role of country banks in the payment system and the first-order effects of payment disruptions for local aggregate demand. This channel bears a resemblance to India’s 2016 demonetization, where [Chodorow-Reich et al. \(2019\)](#) show that districts experiencing more severe cash shortages suffered greater reductions in economic activity in the short run, and is also consistent with the [Friedman and Schwartz \(1963\)](#) hypothesis that the money supply has first-order effects on output.

The source of exogenous variation in financial stress in this setting merits special attention. Several papers have documented the effects of foreign financial crises on the credit supply decisions of domestic lenders ([Bottero et al., 2017](#); [Huber, 2018](#); [Ongena et al., 2015](#)). To the best of the authors’ knowledge, only in one other study ([Acharya et al., 2018](#)) is the case where *foreign* sovereign debt devaluation was the driving force behind deterioration of the domestic banking system’s balance sheet considered, focusing on the European debt crisis. Banks in EU countries contracted lending due to the balance sheet effects of sovereign debt devaluation (see [Bocola 2016](#) for a model of this channel) but the effects were concentrated in banks from the five countries (Greece, Ireland, Italy, Portugal, Spain) most affected by the sovereign debt crisis. The present analysis provides further evidence of this particular cost of sovereign default: contraction of financial intermediation in the investing country, even by domestic banks. The existence of this channel in early 19th century capital markets is a novel finding.

2 Historical Context

2.1 Country Banks in England

The banking system in England during this period was a three-tiered structure comprised of the Bank of England, London banks, and country banks. Country banks began to appear in England in the mid-18th century, around the time the first industrial revolution began. By 1815 there was a country bank within 15 miles of anywhere in England ([Pressnell, 1956](#)).

Each country bank maintained an account with a bank in London, referred to as its London agent. The London agent performed important functions for country banks, transferring excess capital from one part of the country to another, particularly from agricultural areas to industrial areas ([Pressnell, 1956](#)). London agents also settled transactions among different country banks. For these services, country banks compensated their agents by promising to leave a large permanent deposit at their London bank ([Pressnell, 1956](#)). This scheme meant that if a country bank’s London agent failed, as several did during the Panic of 1825, the country bank faced the substantial capital loss of their London balance. In line with recent findings for the U.S.’s correspondent banking network of the 1920s and 1930s (see [Calomiris et al. \(2022\)](#)) and with the broader literature on the relationship between interbank linkages and financial contagion ([Anderson et al. \(2019\)](#) and [Acemoglu et al. \(2015\)](#)), we show that this network was an important source of propagation of financial shocks.

Legal restrictions capped the number of partners in a bank at six, so country banks were small and served a limited geographic area. Country banks took deposits, but these tended to only be from wealthy individuals and large firms. The main assets, other than cash reserves, were loans to local firms and purchases of “London assets”: British government securities and interest-bearing balances with London agents. Country banks tended to match their assets to their liabilities. Private country bank note issues were backed by cash reserves and government securities; deposits were used to discount local or London bills of exchange (similar to commercial paper), and longer term loans came from bankers’ capital ([Pressnell, 1956](#)).

2.2 Exogeneity of the Panic of 1825 to Local Conditions

The key to our identification strategy will be that the Panic of 1825 crisis originated outside the small town economies we study. In this section we first discuss the causes of the crisis, then provide a timeline of events during the crisis, and discuss the conduct of the Bank of England and other policy responses.

The Panic of 1825 has been called the first Latin American debt crisis ([Dawson, 1990](#)). The success of Barings’ French bond offerings in 1820, combined with the prospect to invest in metal-rich newly independent Latin American governments, and the low return on British government consols, created a huge demand for Latin American securities in the early 1820s ([Neal, 1998](#)). As John Horsley Palmer, the Governor of the Bank of England in 1832, put it: “the excitement of that period was further promoted by the acknowledgement of the South American republics by this country, and the inducements held out for engaging in mining operations, and loans to those governments” ([House of Commons Repts. and papers, 722, 1832](#)). These Latin American issuances constituted the “largest single category of new investment” in the lead up to the crisis ([Gayer et al., 1975](#)). [Kenny et al. \(2021\)](#) classify eight financial crises in England from 1750-1938 as endogenous (caused by economic depression or government policy) or exogenous (fraud or risk management related) and find that the Panic of 1825 was exogenous.

Information on the quality of Latin American bonds emerged in London slowly over the years following the first issuances in 1822. Because little was known about each individual government, all Latin American bonds were priced at a heavy discount, and prices for all countries tended to move together before 1825 (see [Figure 1](#)). So little information was available about Latin America at the time that Scottish explorer Gregor McGregor was able to issue bonds for the fictional Central American government of Poyais on similar terms as bonds issued for Chile ([Morgan and Narron, 2015](#)). Latin yields surged during the fall of 1825 as the Poyais and other schemes were revealed. A November 20, 1825 article in [The Examiner \(1825b\)](#) questioned Argentina’s ability to meet its next coupon payment. A December 8 report ([The Morning Chronicle, 1825](#)) noted that “every description of Foreign Security continues under a cloud—but more especially South America.” By January 1826 it

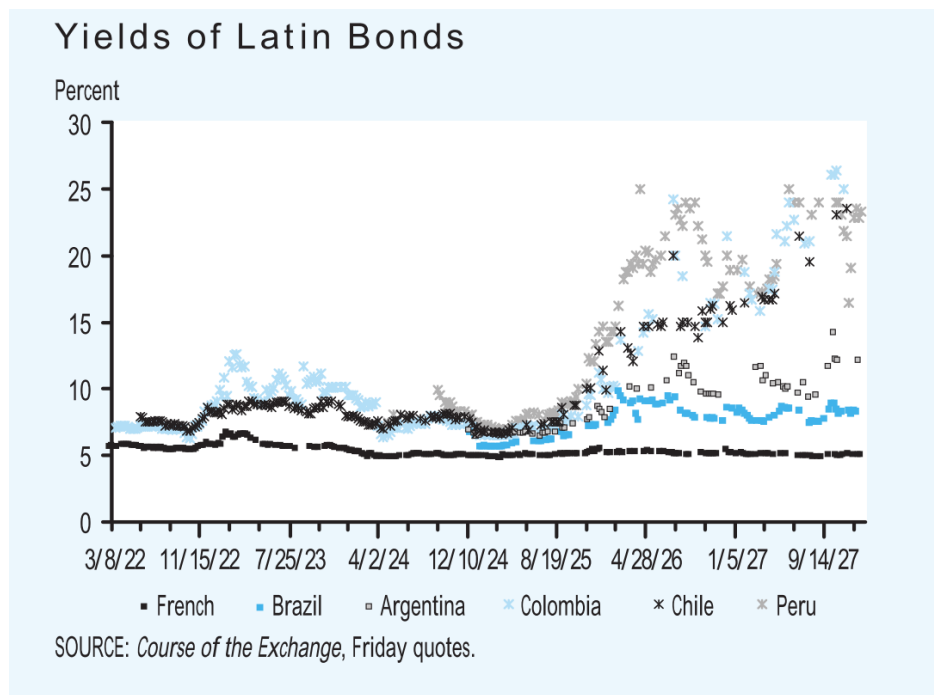


Figure 1: Bond yields in London, 1822-1827. Source: [Neal \(1998\)](#).

was clear that most, if not all, Latin American borrowers were insolvent ([Dawson, 1990](#)), though it was not until April 1826 that the first country, Peru, formally defaulted.

London banks assumed two roles in sovereign debt issuances. First, banks could underwrite foreign government debt, assuming liability in case of default. Second, banks could assume a “payee” role, responsible for collecting the purchase payments of bondholders and then acting as a “window” on behalf of the issuing country and paying out coupon payments.² [Dawson \(1990\)](#) describes a case in 1823 when the London bank Everett & Co., the payee for Peru, temporarily froze payments from loan subscribers to the Peruvian government because of uncertainty about regime stability. Peru needed these payments to make coupon payments back to bondholders and ended up suing Everett & Co. to release the money. Incidents like these undermined confidence in both Peru’s and Everett’s ability to meet their obligations and suggest that payee banks were sometimes expected to supply the coupon payments when the sovereign could not come up with the money.

Rumors of impending defaults put immense pressure on the London banks involved in the

²[Flandreau and Flores \(2009\)](#) make the distinction between underwriting and being the window/payee on sovereign debt in the 1820s. They argue that “the risks and revenues of the last two operations were much smaller than those from the first, but leads and lags could cause trouble.”

debt issuances of these countries. The failures of sovereign borrowers associated with a bank could have devastating reputational consequences, according to [Flandreau and Flores \(2009\)](#) and [Indarte \(2024\)](#). Often the only information published about sovereign debt issuances in the 1820s were the amount of debt, the interest rate, the underwriter, and which bank would make the coupon payments.

Three of the six largest London agents (by number of country bank correspondents) that failed during the crisis had been payees on Latin American debt issuances, ([Dawson, 1990](#)). Perring & Co. paid interest on Poyais, Everett & Co. on Peru, and Fry and Chapman on Mexico and on a portion of the Peruvian debt. These three London agents accounted for nearly half of the country banks exposed to the failure of their London agent during the crisis. Other London banks that had no country bank correspondents but were also involved in underwriting and issuing South American securities also failed during the panic: Goldschmidt failed in February 1826 and Barclay, Herring, Richardson & Co. in April 1826. Underwriting banks often intervened to support securities' prices by buying them up during selloff periods. Holding these assets was costly: Peruvian bonds earned a negative return of 15 percent during this period, for example ([Flandreau and Flores, 2009](#)).

Other causes of the Panic of 1825 have been suggested. [Neal \(1998\)](#) identifies 624 companies floated in England from 1824 to 1825, only 127 of which survived to 1827. Many of these were international companies, particularly mining companies in South America. The Bank of England had also created accommodative monetary conditions since 1819 by lowering its discount rate from 5 percent to 3 percent. [Pressnell \(1956\)](#) argues that a fall in the rate of interest on their London balances encouraged country banks to search for higher yield by increasing their own note issues from December 1823 to December 1825, but also explains that these increases were matched by increases in the demand for credit due to good harvests and increased foreign trade. All three of these causes, the sovereign debt crisis, the decline in foreign private stock prices, and accommodative monetary policy, were more or less external to real economic activity in the interior of England.

A concrete timeline of the crisis helps to clarify our identification strategy. Against the backdrop of increasing tightness in the London money market caused by the sovereign debt crisis, the panic began on December 12, 1825 when the London bank Pole, Thornton &

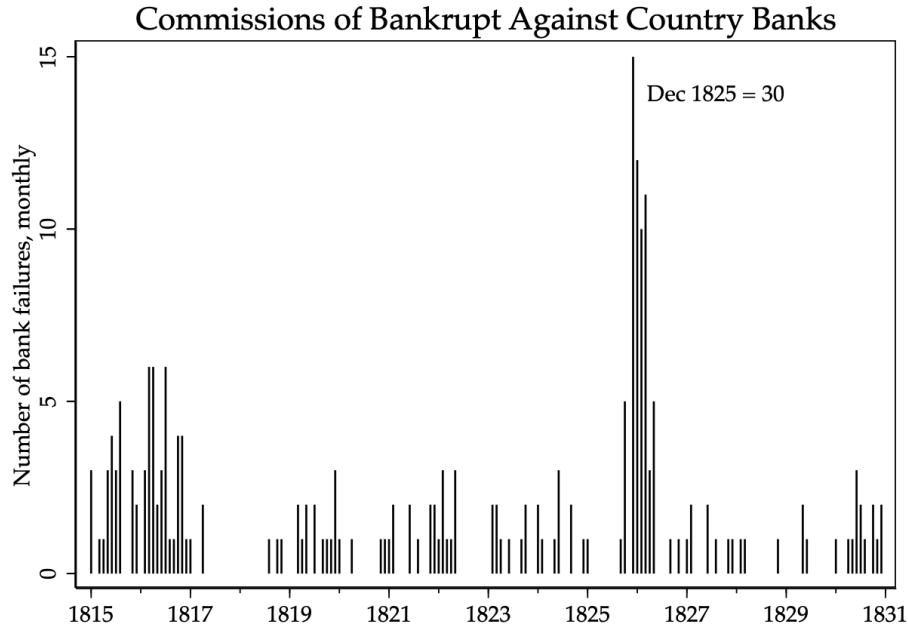


Figure 2: Source: Report from the Committee of Secrecy on the Bank of England Charter, 1832, Appendix No. 101.

Co. stopped payment.³ News of the crisis spread rapidly and runs began the next day on country banks known to be Pole’s correspondents. Runs also occurred on other London banks suspected to be in trouble. By the end of that week four major London banks with a total of 65 country correspondents had failed. By the end of December, 30 country banks had been declared bankrupt and 41 more would follow suit from January to May of 1826 (Figure 2).

James (2012) called the Bank of England’s response to panics from 1790 until 1825 “limited, episodic and inconsistent”, but noted the more active role the Bank played in addressing the 1825 crisis through increased lending. However, the Bank also increased the discount rate from 4% to 5% on December 14, 1825. The Bank also began to issue one pound notes during the crisis as an emergency measure (all small notes had been removed from circulation in 1821). Parliament set up special boards to make advances to economically distressed country towns.

³On the causes of Pole, Thorton & Co.’s failure, [The Examiner](#) (1825a) writes “The decline of this house is generally attributed to the anxiety felt by the partners at the time when the rate of interest was low, to make a profitable use of their capital, and hence they were led to employ it on securities capable of being realized only at a distant period, or of an inferior degree of credit.”

The Panic of 1825 eventually caused significant reforms to the English banking system. In 1826 Parliament lifted the prohibition on joint-stock banking outside London that had been in place since 1708 (Black, 1995). These larger banks slowly absorbed or out-competed the country banks. At the same time, the Bank of England sought to expand its sphere beyond London by opening branches in seven major English cities. Because small note issues were blamed for the crisis, in March 1826 Parliament declared that all private notes below five pounds had to be withdrawn by 1829 (Black, 1995). These additional changes to the banking system after 1826 make it difficult to study the longer term effects of the crisis, so we focus our analysis on the short run effects of the crisis.

3 Data

To undertake this study we developed two novel datasets. One database records the universe of country banks in the United Kingdom, matching each country bank to a London bank (agent). Using this data, we assign each district in England a bank status (at least one failed bank vs all active). We then assign each district a failed agent status (at least one bank with a failed agent vs no banks with a relationship with a failed agent). We use these values to create the independent and instrumental variables in our main estimating equation described in Section 5. The other database includes all bankruptcy records in England for the period January 1, 1820 to January 1, 1831, which we use to create the dependent variable in our main specification.

We aggregate both datasets to the district level using the geographic boundaries provided by Schurer and Higgs (2023). As we do not have population data for the counties of Gloucestershire and Staffordshire, we remove districts in those counties from all analyses. We also exclude districts in London as firms in London did not have to rely on country banks. Additional details on the datasets are provided in Appendix A.

3.1 Banking Network

We construct the banking network characterizing the English financial system from 1820-1830 using five years of data from the Post-Office London Directory, a business directory

Table 1: Post-Office London Directories, English Banks Only

	1820	1823	1825	1827	1830
Number of districts with a country bank	262	279	281	251	236
Number of country banks	408	425	425	370	328
Number of country bank branches	492	528	538	447	429
Number of London agents	57	55	54	48	43
Average bank (branch) failures per year	16	30	58	49	

Source: Post-Office London Directories, 1820, 1823, 1825, 1827, 1830. Average bank (branch) failures denotes the average number of failures per year in the years between the given year and the next year in which data is available.

published by the Postmaster General: 1820, 1823, 1825, 1827⁴, and 1830. We identify country bank branch failures on the basis of their disappearance from the 1827 London Directory relative to 1825. Table 1 describes some of the features of the data. Consistent with narrative evidence about the banking system at the time, the number of country bank branches in England peaked in 1825 and declined afterwards. The greatest number of failures per year came between 1825 and 1827. However, a large number of disappearances occurred between 1827 and 1830, likely driven by other changes like the prohibition of small note issues, the introduction of joint stock banking, and competition from the branches of the Bank of England that opened during this period.

3.2 Firm Bankruptcies

The second set of data we collect is individual bankruptcy statistics from the Edinburgh Gazette.⁵ The records include the date of the announcement of the bankruptcy proceedings to the general public, the bankrupt individual’s name, location of residence, and occupation. To our knowledge, this is the first time these records have been collected at the town level rather than at the national level.

⁴We thank a researcher at Reed College for providing access to the 1827 volume.

⁵Bankruptcy notices for all of Britain had to be printed in the London, Dublin, and Edinburgh editions of the Gazette. For English bankruptcies the Edinburgh Gazette prints the information most readably. However, using Edinburgh Gazette entries means we don’t reliably capture bankruptcies in Scotland and Ireland, so we exclude these countries. See Appendix A.2.2 for further discussion.

Bankruptcy commissions could seize an individual’s assets, determine which creditors would be paid, and how much each creditor would receive. To be eligible for bankruptcy, an individual’s total debt had to exceed 100 pounds, equivalent to around 8000 pounds today, and the individual had to be classified as a trader rather than a professional (Duffy, 1973). These criteria remained fixed over the study period. This means that our data omits gentlemen and professionals like attorneys and doctors, and merchants owing amounts under one hundred pounds. Private businesses were not entitled to limited liability during this period because of the Bubble Act of 1720, so we treat individual and firm bankruptcies as equivalent and refer to bankruptcies as firm failures throughout the paper.

The sample we collect covers January 1, 1820 to January 1, 1831 and includes 6,979 bankruptcies in 312 districts, 283 of which had a country bank branch in 1823. The bankruptcy data is quite sparse, with a mean number of monthly bankruptcies at 0.181 and a comparatively large standard deviation of 0.620. Figure 3 shows the total number of bankruptcies across time, marking our preferred start and end dates for the Panic - December 1825 and February 1827. This timeline covers two spikes in bankruptcies that are well above any other period in the decade.⁶ We categorize each bankruptcy using the occupation of the first listed partner as “Tradable”, “Non-Tradable”, “Construction”, and “Other”.⁷ Bankruptcies increased substantially in 1826 across all occupation classes (see Table 10).

While the bankruptcy data does report banker bankruptcies (which we exclude from the analysis on non-financial firm bankruptcies), they are understated compared with the Post-Office directory data. One reason for this is that each bankruptcy notice lists a single location, where the individual actually lived. In many cases multi-branch banks failed corresponding to just one individual named in bankruptcy proceedings. For this reason, we generally use the bank failure data from the Post-Office directories, though using only bank failures in December 1825 provides a similar estimate of our results in Section 5.

A final concern with the bankruptcy statistics is that many troubled debtors may not appear in the statistics at all due to the inefficacy of bankruptcy commissions during this period.⁸ Hearings on bankruptcy laws conducted in 1818 suggested that in cases of debts

⁶In a robustness check, we use only the first spike to mark the end of the Panic.

⁷We discuss our classification methodology in Appendix A.2.1.

⁸Duffy (1973) argues that “faulty laws and administration encouraged dishonesty and prevented speedy

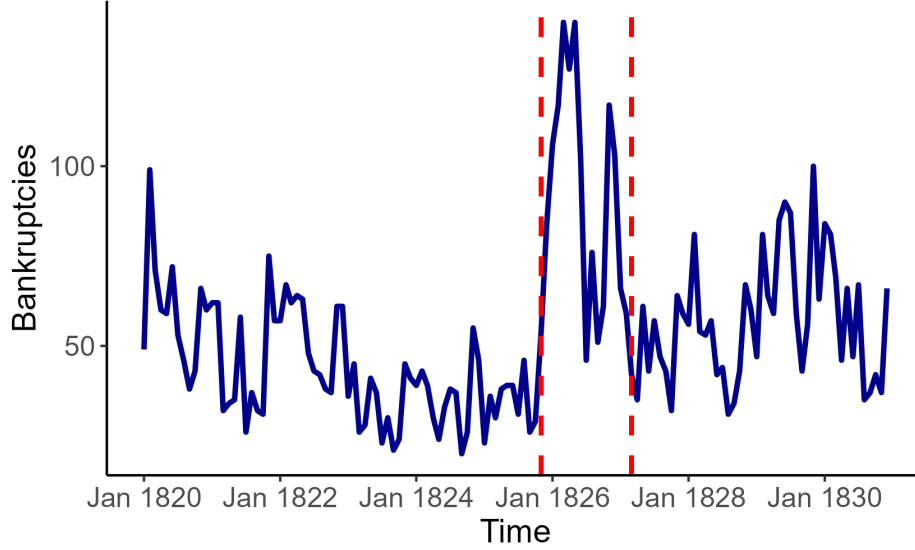


Figure 3: Number of non-financial firm bankruptcies across time. The vertical dotted lines mark our preferred beginning and end months of the Panic - December 1825 and February 1827.

less than £1000 the costs of bankruptcy commissions usually exceeded the amount recovered from bankrupts' estates (Duffy, 1973) and were rarely initiated as a result, so we are likely measuring truly large firms. Because of concerns like these, Silberling (1919) and others have used bankruptcies as a barometer of economic activity rather than a measure of activity in itself. Gayer et al. (1975) show that bankruptcies strongly comove with many other cyclical indicators like trade volumes, indices of goods production, inflation, and the money supply at the national level.

4 Identification Strategy

To estimate the effect of local bank failures on non-financial firm bankruptcies, we begin with the following difference-in-differences specification:

$$f_{it} = \gamma_i + \lambda_t + \beta (BankFailure_i \times Post_t) + \epsilon_{it} \quad (1)$$

where $f_{i,t}$ denotes the number of non-financial firm bankruptcies per 10,000 people in collection of estates" and that bankruptcy laws were unpopular as a result (p. 153).

district i and month t . The difference in difference product $BankFailure_i \times Post_t$ equals one if district i had at least one bank failure between 1825 and 1827, and zero otherwise. The coefficients γ_i and λ_t are district and month fixed effects, respectively, which account for time-invariant district characteristics. Errors are clustered by district to account for potential serial correlation of the residuals across time. We exogenously control for changes in local population (which may drive bankruptcies) by measuring bankruptcies per 10,000 people as the dependent variable, rather than including population as an independent variable.

4.1 Endogeneity Concerns

Although the difference-in-differences specification in equation (1) includes district and time fixed effects, several sources of endogeneity remain that may bias the OLS estimates of β . First, there is a risk of omitted variable bias. Local economic shocks may simultaneously increase the likelihood of bank failure and lead to firm bankruptcies. Time varying shocks are not captured by the fixed effects terms, and the coefficient on $BankFailure_i \times Post_t$ will conflate the effect of bank failure with that of broader unobserved local distress. Second, there might be a simultaneity issue in measuring the relationship between bank failures and firm bankruptcies. The estimation of equation (1) requires that bank failures affect firms, but not vice versa. However, firms defaulting on their obligations could precipitate the failure of their local bank. If it is indeed the case that firm distress causes bank failure, the OLS estimate would be invalid.

4.2 Instrumental Variable Strategy

Our main strategy to overcome these issues is to use an instrumental variable. The instrument leverages the structure of the English correspondent banking network. Each country bank maintained an account with a London bank, its agent, which settled transactions and held a portion of the country bank's reserves. When a London agent failed during the Panic, its country bank correspondents suffered sudden losses to liquidity and capital that could cause the country bank itself to fail. We respectively name the difference in difference variable and the instrument as:

$$BankExposure_{i,t} = BankFailure_i \times Post_t \quad (2)$$

$$AgentExposure_{i,t} = AgentFailure_i \times Post_t \quad (3)$$

where $AgentFailure_i$ equals one if a district had at least one country bank linked to a London agent that failed during the crisis (1825-1827). We then estimate the first stage of a Two Stage Least Squares (2SLS) model by regressing $BankExposure_{i,t}$ on $AgentExposure_{i,t}$ and using the estimated $\widehat{BankExposure_{i,t}}$ as the main independent variable of interest in the second stage:

$$BankExposure_{i,t} = \gamma_i + \lambda_t + \pi AgentExposure_{i,t} + \nu_{i,t} \quad (4)$$

$$f_{i,t} = \gamma_i + \lambda_t + \beta \widehat{BankExposure_{i,t}} + \varepsilon_{i,t} \quad (5)$$

This instrument shifts only the likelihood of a bank failure through exposure to a failed agent, plausibly independent of underlying local economic fundamentals as agents were operated exclusively in London (which we otherwise exclude from our dataset) and were subjected

Table 2: Instrumental Variable First Stage Results

	(1)	(2)	(3)	(4)
<i>AgentExposure</i>	0.712*** (0.043)	0.791*** (0.043)	0.225*** (0.058)	0.225*** (0.069)
Obs.	18500	18500	18500	18500
Districts	250	250	250	250
Mean of Dep. Var.	0.131	0.131	0.131	0.131
Adjusted R-Squared	0.304	0.397	0.688	0.688
F-Stat	272.381	343.444	14.968	10.691
District FE		Y	Y	Y
Month-Year FE			Y	Y
Cluster Unit	District	District	District	County

Notes: Estimation of equation 4 where the independent variable *AgentExposure* is an indicator equal to 1 for all months after November 1825 for districts which have at least one bank with an agent which failed between 1825-1827. The dependent variable is *BankExposure*, an indicator equal to 1 for all months after November 1825 for districts which have at least one bank which failed between 1825-1827.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

to a shock to their holdings of foreign debt.

4.2.1 Relevance: Agent Failures Affected Country Banks

The first condition for instrumental variable validity is relevance: the failure of a country bank’s London agent needs to have had a material effect on the chance that the country bank itself would fail. The first stage results are reported in Table 2 and the instrument appears to be strong. Our preferred estimate, which we use in our main 2SLS results, is in column 3 and includes district and month-year fixed effects and clusters errors by district. To supplement this finding, this section presents bank-branch level regressions (as opposed to district level as in our first stage estimates) to understand the drivers of bank failures and shows that agent failures were an important driver of bank failures. Table 3 shows probit model regression estimates on the cross section of our bank network dataset⁹ for bank failures for 538 English bank branches.

Across all specifications, agent bankruptcy has the expected positive association with bank bankruptcy. Banks with multiple branches were no less likely to fail than unit banks, despite insuring themselves across space. Banks that were founded within the five years before the crisis seem more likely to fail. The number of banks that failed and that had the same London agent as bank i is not predictive, likely because it is strongly correlated with agent bankruptcy. Column (3) provides evidence that the financial crisis was propagated not just through agent failures, but also through a contagion effect. Banks in districts where other banks failed were more likely to fail themselves. This is true even when including county fixed effects.¹⁰

The magnitude of the effect of an agent failure is large. Using the model in column (4) of Table 3 at the mean of the other covariates, branches whose London agent did not fail had an 18% chance of failing during the crisis while banks whose agent failed had a 38% chance. Given this, plus the finding that there was a within-district contagion effect, the first stage regression of district level bank failures on district exposure to London agent failures is not expected to be a weak instrument.

⁹We use the cross section to measure the effect of several time invariant variables.

¹⁰Eight of the thirty nine counties included in our dataset experienced no bank failures, so including county fixed effects omits banks in those counties, thus decreasing the number of observations.

Table 3: Probit Models for Bank Failure

	(1)	(2)	(3)	(4)	(5)
Bank Failure					
Agent bankruptcy	0.652*** (0.137)	0.627*** (0.141)	0.634*** (0.144)	0.684*** (0.197)	0.933*** (0.205)
Number of other branches		0.034 (0.032)	0.025 (0.033)	0.043 (0.035)	0.045 (0.043)
Founded 1821-1825		0.159 (0.136)	0.207 (0.141)	0.223 (0.142)	0.312** (0.158)
Other Bank failures in same district			1.309*** (0.198)	1.336*** (0.200)	1.267*** (0.234)
Agent's number of clients				-0.006 (0.007)	0.000 (0.008)
Bank failures of same agent				-0.009 (0.023)	-0.035 (0.024)
County FE					Yes
Observations	538	538	538	538	488

Note: Country bank branch level probit model where the dependent variable is equal to 1 if country bank branch i fails during the Panic. For both the country bank and the agent variables, failure is determined through specific dates listed in the *Edinburgh Gazette*. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Pressnell \(1956\)](#) uses surviving bank balance sheet data to argue that the London account was the most important part of a typical country bank's balance sheet and the first resource in times of liquidity crisis; across nine country banks, the mean share of total assets held by the London agent was 22%. The banking network data also shows that bank-agent relationships in the English banking system during the 1820s were sticky: 76% of country banks that survived until 1830 had the same London agent as they did in 1820. Switching London agents was likely costly and problems at the London bank were therefore transmitted to the country bank.

4.2.2 Exclusion Restriction

The second requirement for the instrument to be valid is the exclusion restriction: that failures of the London agents serving a town's country banks did not affect local economic conditions, especially firm bankruptcies, in any way other than through financial stress on

Table 4: Regression of Bankruptcies on Bank Failures and Agent Failures

	(1)	(2)	(3)	(4)
Agent Fail	0.090 (0.092)	-0.132 (0.178)	0.033 (0.085)	-0.025 (0.181)
Bank Fail	0.148** (0.068)	0.371** (0.157)	0.120** (0.059)	0.137 (0.128)
Obs.	710	710	710	710
F Stat.	2.796	3.080	20.481	27.643
Mean Dep. Var.	0.370	1.407	0.370	1.407
Included Bankruptcies	Single Year	Cum.	Single Year	Cum.
Fixed Effects			Y	Y

Notes: OLS estimation where the independent variable Agent Fail is an indicator equal to 1 if a district has at least one bank with a London agent failure in the time period and zero otherwise and Bank Fail equals 1 if a district has at least one bank failure in the time period and zero otherwise. The dependent variable is the number of bankruptcies per 10,000 people in the relevant time period. We use years for which we have complete bank network data and population data: 1823, 1825, and 1827. In columns (1) and (3) we use bankruptcies only for those years to calculate the dependent variable; in columns (2) and (4) we sum all non-bank bankruptcy data between bank network years: 1821-1823, 1824-1825, and 1826-1827. Only districts with a bank in 1823 and at least one bankruptcy in any year are included. Fixed effects are included in columns (3) and (4) and are for district and year. Errors are clustered by district.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

the town's country banks. There is little evidence that London agents had a direct effect on districts' economies. Agents only occasionally engaged with firms outside London. It is possible that they may have been more likely to lend to firms in towns where they had a country bank client. Few balance sheets have survived to shed light on this concern. [Duffy \(1973\)](#) collects the claims of major claimants against Brickwood & Co., a London agent that failed in an earlier banking crisis in 1810. For this particular bank, with liabilities of £621,117, only 6% of those were owed to traders outside of London, and just three individuals made up these claims (p. 381). The country bank with the largest balance at Brickwood, Bowles Bank, accounted for about 20% of all outstanding claims. Table 4 shows that banks with an agent failure generally did not affect local bankruptcies in a district, unlike local bank failures which are associated with a positive increase in bankruptcies. Our estimates include data for banks and agents which failed between 1820-1823, 1823-1825, 1825-1827.

4.2.3 No Selection on London Agents

Instrumental variable validity requires that banks with London agents that failed were not systematically different from other banks; the agent-country bank relationship must be plausibly randomized. There is a risk that bad firms build relationships together. In such a scenario, we might expect bad country banks to form relationships with firms that are more likely to default. If the agent-country bank relationship is not randomly assigned, there is the possibility that agents failures are driven by bankruptcies in these bad non-financial firms downstream. Our estimation strategy deals with some of these concerns and we see no pre-trends in Figure 4 that might indicate that something other than the Latin American crisis is causing this increase in bankruptcies. Random assignment of agents would preclude additional forms of potential endogeneity.

Not much is known about how agents were chosen by country banks. Pressnell argues that the choice of a particular London banker was affected largely by the nature of the business of the country banker and of his clients, but family ties also played a role. The fact that relationships were so sticky, as already demonstrated, makes it unlikely that more savvy banks were able to foresee and avert risks related to which London agent they used.¹¹

Table 11 (located in the appendix) compares banks with failed London agents to those whose agent survived and shows few differences. Banks in these two groups were equally likely to have more than one London agent, be founded in the last five years, had roughly the same number of competitors in their districts, and their districts had roughly the same number of firm bankruptcies in the pre-crisis period. The only statistically significant differences between the two groups are in the number of bank branches, with exposed banks having more bank branches on average, and exposed banks tending to be in towns with lower populations. In the regression analysis we use per capita firm bankruptcies as our dependent variable to account for this latter point.

Historical evidence also supports the fact that the exposed banks were no more risky than other banks ex ante. Many country bank failures during the 1825 Panic were caused by illiquidity rather than insolvency. By 1828 23 out of 63 banks that declared bankruptcy

¹¹We also show in a placebo test in Section 6 that having a London agent that failed in 1825 did not predict bank failure in 1823-1825.

during the crisis had resumed payment, and records from the same year show that an additional 31 of these 63 were still attempting to resume operation (Pressnell 1956 p. 491).¹² Still, payment stoppages that were resolved years later could have large consequences in the short run (Section 5.4).

Maps in Appendix A.5 show that districts with failed agents did not generally clump together. This is also true at the county level where the number of failed agent with a banking relationship in a county is closely related to the total number of unique agents with a banking relationship in the county. The number of agent failures per bank also does not appear to exhibit geographic clustering. Districts which suffered both bank failures and agent failures during the Panic were spread throughout the country. These findings suggest that bad agents did not exhibit a geographic pattern in terms of their relationships with country banks and it is unlikely that there was geographic selection whereby bad agents entered markets that had relatively poor economic outlooks or worse banks.

5 Results

5.1 OLS Results

Figure 4 shows an event study plot based on equation (1). The period preceding the Panic does not show any apparent trends in firm bankruptcies following Panic-induced bank failures. The comparison is between districts which experience a bank failure and districts which do not, and as the pre-Panic coefficients are all essentially zero, the parallel trends assumption likely holds. Though no single month in the post-period is statistically distinct from zero, the aggregate post-period coefficient is, potentially because the data are noisy due to sparsity. Aggregating at the quarterly level confirms this; the estimate for the second quarter of 1826 - the height of the crisis - is greater than zero and statistically significant.

We estimate the OLS model described by equation (1) and present the results in column (1) of Table 5. We restrict our sample to districts with at least one bank in 1823 so that we

¹²Surviving bankruptcy records from three of these banks show that two were solvent and the third was short only £6,000 on a debt of £71,000 (Pressnell, 1956).

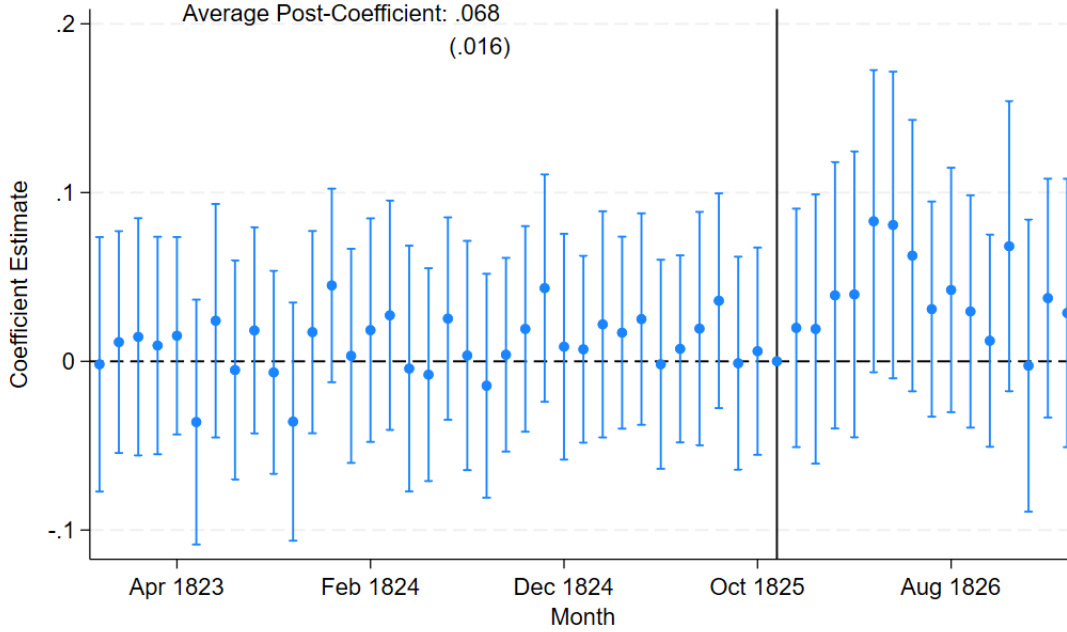


Figure 4: Event study plot based on estimation of (1) where the main independent variable is the interaction between month and an indicator equal to one if a district has at least one country bank that fails between 1825-1827 and the dependent variable is the number of firm bankruptcies. November 1825 is the comparison month. The average post-period coefficient and error term are in the top left.

are only comparing districts with access to the financial markets.¹³ Additionally, we restrict our analysis to districts with at least one bankruptcy in the time period of study: January 1821¹⁴ - February 1827.

The coefficient estimate of .028 is statistically significant, indicating a positive relationship between bank failures and non-financial firm failures. Districts with a bank failure between 1825-1827 are associated with an increase of .028 firm bankruptcies per 10,000 people compared with districts without a bank failure. While this number may appear small, as discussed in Section 3 our dataset undercounts firm bankruptcies; we view the positive relationship as an indicator that bank failures are positively correlated with poor economic conditions.

¹³We use 1823 instead of 1825 as the year with which we establish a district's access to the financial network in case banks had already failed during the Panic when they were recorded in the 1825 Post-Office Directory. A robustness test which uses 1825 instead of 1823 indicates that the choice between the two is largely irrelevant.

¹⁴We begin the analysis in 1821 rather than 1820 due to missing data on districts' population levels prior to 1821.

Table 5: Main Results - Effect of Bank Failure on Firm Bankruptcies

	(1)	(2)	(3)	(4)	(5)
<i>BankExposure</i>	0.028*** (0.010)	0.138** (0.061)	0.235** (0.103)		
<i>AgentExposure</i>				0.031** (0.012)	0.031* (0.016)
Obs.	18500	18500	16500	18500	18500
Districts	250	250	250	250	255
End Month	Feb 1827	Feb 1827	June 1826	Feb 1827	Feb 1827
Mean	0.058	0.058	0.056	0.058	.059
Analysis	OLS	2SLS	2SLS	Reduced Form	SDID
Kleibergen-Paap		14.969	14.969		

Notes: Column (1) reports the coefficient estimate of the OLS model described by equation (1). Columns (2) and (3) report coefficient estimates of the 2SLS model described by equation (5) which includes district and monthly fixed effects. The dependent variable is the number of non-financial firm bankruptcies in district i and month t . Column (3) uses an earlier, alternate date for the end of the Panic. Column (4) estimates the reduced form model where the main independent variable is an indicator equal to 1 if a district contains a bank whose agent failed between 1825-1827. Column (5) reports the estimate of the reduced form Synthetic Difference in Difference model. Errors are clustered by district in columns (1)-(4) and bootstrapped in column (5). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.2 2SLS Results

The 2SLS coefficient estimate of .138 (column 2 of Table 5) is statistically significant and implies a large effect on bankruptcies; districts with country bank failures due to exposure to a failed London agent experience an average increase of .138 firm bankruptcies per 10,000 people per month during the Panic, relative to districts without agent exposure. Seen another way, firm bankruptcies in districts with bank failures increase by 237% compared with districts that do not experience agent-driven bank failures.¹⁵ The total number of firm failures caused by each bank failure might appear modest, but, as discussed in Section 3, we view bankruptcies as a barometer of overall local economic activity. The Kleibergen-Paap F-statistic of 14.97 indicates that the instrument is unlikely to be weak.

The coefficient estimate is almost five times larger than the OLS estimate of .028 in

¹⁵We approximate the increase by dividing the coefficient of .138 by the mean of the dependent variable of .058.

column (1). The 2SLS estimate is even larger when using an alternate, earlier cutoff for the end of the Panic (column 3). A likely reason for the difference in magnitude with OLS is because the local average treatment effect (LATE) is much larger than the average treatment effect (ATE). The reduced form IV estimate is almost identical to the OLS estimate, suggesting that this is indeed the case, rather than other possible explanations such as the instrument causing a sizable reduction in measurement error or in omitted variable bias. One explanation for the difference between LATE and ATE is that the districts which experienced an agent failure experienced a larger effect from a bank failure. Figure 5 shows the event study plot including only districts which had a bank failure in the post-Panic period (between 1825-1827) and where districts which experienced an agent failure are considered treated (and those without an agent failure are the control). Though there is no difference in the pre-period, the first few months after the start of the Panic sees an increase in the coefficient indicating that there is indeed a larger effect in agent-failure districts. This also points to a possible contagion effect (which we discuss below) having a weaker effect on firm bankruptcies than the agent-driven failures. The OLS estimate includes both the initial shock and the subsequent contagion effect, but as the shock is driven by agent failures, the 2SLS effect will skew heavily towards the initial shock. An immediate (agent-failure driven) shock presumably gives firms no time to adjust and prepare for a downturn, causing a larger effect on firm bankruptcies. This is reflected by the difference in columns (2) and (3); since column (3) uses an earlier cutoff date it includes less of the contagion phase of the crisis and the coefficient is much larger than in column (2).

Figure 4 show that there is no pattern in the pre-trends in the OLS model, indicating that the parallel trends assumption likely holds. To further establish parallel trends, we estimate the reduced form model using the Synthetic Difference-in-Difference (SDID) estimation strategy proposed by Arkhangelsky et al. (2021). The results are almost identical to the reduced form estimates in column (4). The event study plot produced by SDID (in the Appendix) is noisy, likely due to the sparsity of the dependent variable. A version in which we aggregate the time element to the quarterly level produces smoother pre-trends which appear parallel.

To understand the aggregate implications of bank failures during the crisis, we perform

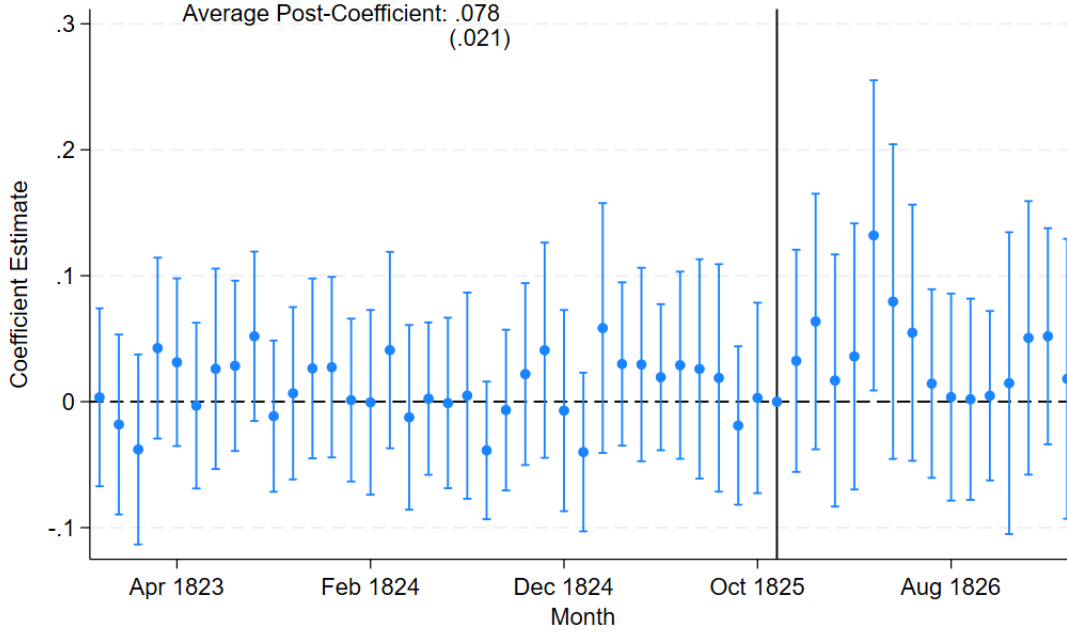


Figure 5: Event study plot based on estimation of (1) where the main independent variable is the interaction between month and an indicator equal to one if a district has at least one country bank with a relationship with an agent that fails between 1825-1827 and the dependent variable is the number of firm bankruptcies. Only districts with at least one bank failure are included. November 1825 is the comparison month. The average post-period coefficient and error term are in the top left.

a simple back of the envelope calculation. Multiplying the number of districts with a bank failure by the implied elasticity of 2.37 (derived from column 2 of Table 5) and by the number of bankruptcies per 10,000 people in 1826 in districts without a bank failure, we find that there were 25.7 additional bankruptcies per 10,000 people than in districts without a bank failure (with a standard deviation of 6.6).

5.3 Contagion

We have shown how the sudden collapse of several London agents served as an immediate shock to those agents' country bank partners. We now study the effects of the financial contagion, the effects which were not driven by agent failures.

5.3.1 Second Wave of the Crisis

Figure 5 shows the event study plot comparing districts with agent-driven bank failures and non-agent driven bank failures. The measurement is noisy, but 7 months after the start of the Panic, the coefficient returns to zero, indicating that the agent failure channel only caused a spike in bankruptcies beyond what other districts with bank failures experienced until June 1826. This suggests that the second spike in bankruptcies - in the latter half of 1826 - is not driven by agent failures, but perhaps by financial contagion.

There are 22 districts that had their greatest number of bankruptcies over a 6 month period during the second half of 1826. These districts are driving the second spike in bankruptcies during 1826 seen in Figure 3 and are likely to be the districts most affected by contagion following the shock caused by the agent failures. Compared with the 45 districts that experienced the greatest number of bankruptcies in the first half of 1826, these “Second Wave” districts have a roughly similar share of bank failures (77.3% in the Second Wave districts vs 82.2% in the initial spike districts) but a lower level of agent failures (40.9% vs 51.1%), perhaps reflecting that agent failure is less predictive of economic downturns in these districts.

As our focus is now on what occurred after the initial agent failure period, our instrument is no longer valid and we present correlational evidence through OLS estimation. Table 6 provides evidence that the Second Wave districts experience the greatest increase in bankruptcies in the second half of 1826. Column (1) provides the baseline estimate of the OLS model and column (2) includes the interaction between *BankExposure* and a indicator equal to one if a district is a Second Wave district. The relatively large, positive coefficient on the interaction term indicates that the Second Wave districts did indeed experience a larger increase in bankruptcies following a bank failure compared against all other districts. In columns (3) and (4) we modify *BankExposure* to equal one for bank failure districts from the starting point of the second wave (June 1826), rather than from the start of the Panic. The overall magnitude of the coefficient is smaller and with a larger relative error, as expected as we do not consider the initial phase of the crisis as part of the pre-period in this specification. However, the interaction term is much larger in column (4) than in column (2), again indicating that the Second Wave districts see a larger increase in firm bankruptcies

Table 6: Effect of Bank Failures on Bankruptcies During Second Wave of Crisis

	(1)	(2)	(3)	(4)	(5)
<i>BankExposure</i>	0.028*** (0.010)	0.021** (0.010)	0.019* (0.010)	0.005 (0.009)	0.045 (0.051)
<i>BankExposure</i> X 2nd Wave		0.063*** (0.020)		0.150*** (0.026)	
Model	OLS	OLS	OLS	OLS	2SLS
Post Start	Dec 1825	Dec 1825	June 1826	June 1826	June 1826
Obs.	18500	18500	18500	18500	18500
Districts	250	250	250	250	250
Mean	0.058	0.058	0.058	0.058	0.058
Kleibergen-Paap					14.969

Notes: OLS estimation of equation (1) which includes district and monthly fixed effects. The dependent variable is the number of non-financial firm bankruptcies in district i and month t . The main independent variable is defined by equation (2) for columns (1) and (2) and modified for columns (3)-(5) such that *BankExposure* is equal to 1 for all months from June 1826 until the end of the period (February 1827) for districts with a bank failure between 1825-1827. “2nd Wave” is an indicator equal to 1 if district i had its greatest 6 month period of bankruptcies between July 1826 and December 1826 and zero otherwise. Column (5) estimates the 2SLS model using agent failures as the instrument. Errors are clustered by district in all columns. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

during the latter half of 1826 than during the initial phase. When using this later start date to demarcate the post treatment variable, our 2SLS estimate of *BankExposure* is no longer statistically significant (column 5), further reinforcing the fact that only the initial shock is driven by London agent failures.

5.3.2 Bankless Districts

In Table 7, we present evidence that bankless districts were largely insulated from the effects of the financial crisis. We estimate the 2SLS model in column (1), but unlike in our main specification, we include districts with no banks so that the comparison is between a treated group of districts with an agent-driven bank failure and all other districts. The coefficient estimate is smaller than in our main specification though still positive and strongly significant. The OLS estimate is unchanged from our main specification. Taken together, these results suggest that bankless districts behaved similarly to districts which had a bank,

Table 7: Estimation Results by District Bank Status

	(1)	(2)	(3)	(4)	(5)	(6)
<i>BankExposure</i>	0.065*** (0.021)	0.028*** (0.008)	0.030*** (0.009)	0.021** (0.008)	0.004 (0.008)	0.004 (0.008)
Model	2SLS	OLS	OLS	OLS	OLS	OLS
Treated	Bank Fail	Bank Fail	Bank Fail	Bank Fail	No Fail	No Fail
Control	No Fail + Bankless	No Fail + Bankless	Bankless	Bankless	Bankless	Bankless
Post Start	Dec 1825	Dec 1825	Dec 1825	June 1826	Dec 1825	June 1826
Obs.	31672	31672	25160	25160	19092	19092
Districts	428	428	340	340	258	258
Mean	0.047	0.047	0.049	0.049	0.035	0.035

Notes: Results of 2SLS estimation of equation (5) in column (1) and OLS estimation of equation (1) in columns (2)-(6). *BankExposure* is defined in equation (2) and equals one if district i has a bank failure from the Post Start date onwards and zero otherwise in columns (1)-(4) and it equals 1 if a district has a bank but no bank failures in columns (5) and (6) and zero otherwise. Columns (1) and (2) include all districts; columns (3) and (4) exclude districts with at least one bank and zero bank failures between 1825-1827; columns (5) and (6) exclude districts with a bank failure between 1825-1827. Errors are clustered by district. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

but no bank failure during the crisis (the control group in our main specification).

In columns (3) and (4) we estimate the OLS model excluding districts with banks that did not fail during the crisis. The results are quite similar to the baseline OLS estimate in column (2). Using December 1825 or June 1826 to define the post period of our *BankExposure* variable does not appear to matter, suggesting that the bankless districts were insulated from both phases of the crisis.

In columns (5) and (6) we consider districts which contained at least one bank and which did not experience a bank failure as the treated group and the bankless as the control group. The coefficient estimate is essentially zero and insignificant, further indicating that the bankless districts behaved like districts with banks which did not fail during the crisis.

5.3.3 Trading Partners

We next explore how a district's trading partners contribute to the contagion effect. We identify a district's top trading partners using market access data from Ravalli (2025), where district i market access is calculated as the sum across all districts j of the product of district

j 's local GDP in 1815 and the cost of commuting between districts i and j :

$$MA_i = \sum_{j=1}^J GDP_{j,1815} * c_{ij}^{-\theta} \quad (6)$$

$$c_{ij} = distance_{ij} * cost_{ij}$$

The cost c_{ij} remains fixed across time and so district j 's contribution to the market access of district i is only a function of distance, market size, and proximity to the canal network. The value for the trade elasticity θ is taken from [Donaldson and Hornbeck \(2016\)](#) and is set at 8.22. To determine the importance of a trading partner, we rank each district j by its contribution to district i 's total market access. We define a top trading partner as a district in the top 3 by rank.

Using several interaction terms involving districts' top trading partners, we estimate our baseline OLS model and report the results in Table 8. We find that a district's top trading partners exerted little influence during the second phase of the crisis. Columns (1) and (2) report the coefficients of *PostXPartnerFail* where *Post* is equal to one for all months after June 1826 and zero otherwise and *PartnerFail* is equal to one if any of a district's top trading partners had a bank failure (column 1) or agent-driven bank failure (column 2). Columns (3) and (4) include an additional interaction with the variable *BankFail* which is equal to one if district i suffered a bank failure and zero otherwise. Only the columns with the agent-driven bank failure term are distinct from zero, though not statistically significant for a p-value less than .05. That the agent driven failures in trading partners appear to have a positive effect on local bankruptcies while other bank failures do not further points to the larger impact that the London agent failures had during this period; districts with agent-driven failures had a greater increase in bankruptcies in the initial phase of the crisis (as seen in Figure 5 and discussed above) and, unlike the non-agent driven bank failures, this increase may have been large enough to trigger firm bankruptcies in trading partners.

We also find (in columns 5 and 6) that, while the lagged firm bankruptcies in district i 's trading partners' is positively correlated with district i firm bankruptcies (indicating that

Table 8: Contagion Through Trading Partners

	(1)	(2)	(3)	(4)	(5)	(6)
Post X Partner Fail	-0.006 (0.012)	0.019* (0.011)				
Post X Bank Fail X Partner Fail			-0.004 (0.017)	0.025 (0.015)		
Partner Bankruptcies					0.119** (0.051)	0.098** (0.048)
Post X Partner Bankruptcies					-0.058 (0.071)	
Post X Bank Fail X Partner Bankruptcies						0.074 (0.085)
Partner Fail Type	Bank	Agent	Bank	Agent	Firm	Firm
Post Start	June 1826	June 1826	June 1826	June 1826	June 1826	June 1826
Obs.	18500	18500	18500	18500	5750	5750
Districts	250	250	250	250	250	250
Mean	0.058	0.058	0.058	0.058	0.173	0.173
Period	Monthly	Monthly	Monthly	Monthly	Quarterly	Quarterly

Notes: The dependent variable is the number of firm bankruptcies per 10,000 people in district i period t . Each column includes district and period fixed effects. “Post” is equal to one for all periods starting in June 1826 and zero otherwise. “Partner Fail” is equal to one if any of district i ’s top trading partners suffered a bank or agent driven bank failure and zero otherwise. “Bank Fail” is equal to one if district i suffered a bank failure. “Partner Bankruptcies” is the mean lagged quarterly bankruptcies per capita averaged across the trading partners. Errors are clustered by district. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

these economies are likely linked), there is no additional effect during the crisis.¹⁶ This result holds even when conditioning on district i bank failure status.

¹⁶Due to the sparsity in the firm bankruptcy data, we aggregate this specification to the quarterly level.

5.4 Discussion

The above results show a causal relationship between bank failures and firm failures in the short run, suggesting a role for banks in promoting economic activity in the towns they served. In this section we describe two possible ways for bank failures to cause firm failures: an aggregate demand mechanism by which the failure of a bank caused a general economic downturn that reduced consumption spending and a credit supply shock mechanism by which banks lent less money to firms.

5.4.1 Aggregate Demand

Towns outside London did not usually use Bank of England notes until after 1826, and instead relied on country bank notes.¹⁷ Note issues were 34% of all liabilities for 12 banks for which we have data between 1790-1826 ([Pressnell, 1956](#)). Day laborers' wages were usually paid with country bank notes and these notes subsequently circulated through purchases of local goods and services. Failures of the banks backing these notes could wipe out the household wealth of noteholders as well as wealthier depositors. For example, the failure of the bank Turner, Turner, & Morris, was said to have caused "much alarm and difficulty among the middling and lower orders, as the circulation of their notes was very great" ([The Examiner, 1825a](#)).

Whether the value of bank notes issued by bankrupt bankers was wiped out entirely or whether they continued to circulate at a fraction of their face value is unclear and seems to have varied. There were some cases where a particular local merchant would accept bank notes of a defunct bank in exchange for goods at a fraction of their face value, hoping to recover some of the value in bankruptcy proceedings according to parliamentary testimony ([House of Commons Reports and Papers, 1848](#)).¹⁸ At other times, small noteholders themselves were forced to participate in bankruptcy proceedings to recover the value. Banks

¹⁷Testimony given to Parliament by the Governor of the Bank of England in 1832 suggested that the public had preferred private notes issued by a banker they knew and trusted to notes issued by the Bank of England that were subject to forgery to a greater degree than country notes ([House of Commons Repts. and papers, 722, 1832](#)).

¹⁸A notice posted in the [Northampton Mercury \(1826\)](#) asserted that anyone who purchased or accepted bank notes of already bankrupt banks had no legal claim to recover the value in bankruptcy proceedings, though the legality of this behavior one way or the other is unclear.

that failed in 1825 eventually paid an average of 85% on their obligations (a figure of 17 shillings on the pound, worth 20 shillings, was cited by [House of Commons Reports and Papers \(1848\)](#).) However, this dividend on the bankrupt's estate was paid out several years after payment was stopped ([Duffy, 1973](#)). Whether the noteholder received a large fraction of the value in some years' time, a smaller fraction immediately, or nothing at all, all three constitute a drop in household money balances in the short run.

5.4.2 Credit Supply Shock

The devaluation of existing bank notes held by households was part of a larger drop in the availability of country bank notes that were the primary means of payment in British towns. Banks supplied their bank notes to firms as a form of working capital in exchange for longer term promissory notes. Pressnell writes that “the bankrupted banks represented a reduction of the means of payment and an immobilization of much capital. The survivors contracted their lending. . . contraction enforced by caution was reinforced by reduced confidence in the ordinary banks” (p. 491). Thus, much like the 2008 financial crisis, the Panic of 1825 was characterized by a contraction in bank lending to firms. [James et al. \(2013\)](#), studying similar disruptions to the payment system in correspondent banking networks in the U.S., argue that these stoppages act as severe adverse supply shocks, mainly by preventing firms from being able to make payroll. They find that payment stoppages by New York banks (analogous to London banks) from 1866-1914 were associated with declines in real activity of 10-20%.

Figure 6 approximates the contraction in short term lending by showing the volume of *new* bank notes issued, differentiating small and large denominated bills. Small denominated bills were commonly used to pay workers' wages.¹⁹ The Bank of England responded to the credit crunch by issuing its own notes²⁰, but not enough to completely offset the contraction in private notes. The fact that small private bank note issues never recovered was due to the 1826 banking reform requiring that notes under £5 had to be withdrawn by 1829.

¹⁹The cash in advance model of [Sargent and Velde \(2002\)](#) provides an example of how the composition of the money supply can play a role in determining output. Smaller denominations provide greater liquidity services in the model.

²⁰[The Examiner \(1825a\)](#) quotes a local Birmingham paper: “the failure of the house of Smith and Gibbins created a good deal of local inconvenience from the quantity of their paper which was in circulation...It appears that the issue of £1 Bank of England notes in Birmingham, has been very considerable, and by no means unwelcome.”

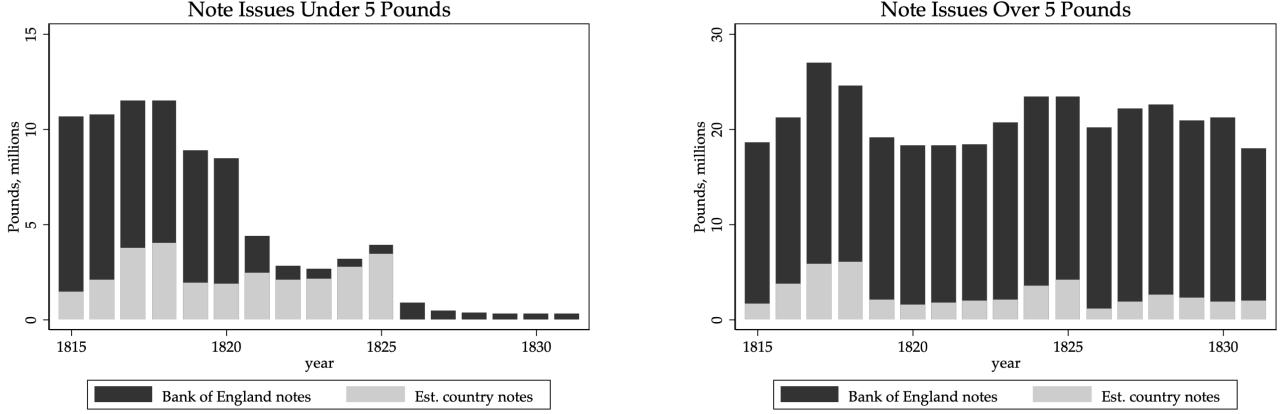


Figure 6: Source: Report from the Committee of Secrecy on the Bank of England Charter, Appendix No. 99 (stamp duties) and Appendix No. 82 (Bank of England notes). We estimate the volume of country bank notes using tax rates and the amount of stamp duties collected.

Unfortunately, no data on note issues at the local level is available so it's not possible to identify the causal effect of credit contractions on firm failures during this period.

The above discussion provides support for the arguments of [Crouzet \(1972\)](#), [Pollard \(1964\)](#), and [Mathias \(1973\)](#) that country banks did indeed contribute, at least modestly, to industrialization, but largely not through long-term lending for fixed capital formation, as [Brunt \(2006\)](#) argued more recently. The above reexamination of the historical evidence suggests that country banks provided liquidity to provincial economies and greased the wheels of nascent factory systems by supplying a means of payment through working capital loans.²¹ This is why we can detect effects of country bank failures in the short run period during and after the crisis. If long term lending were the main driver, so many failures of large firms would likely not have occurred so rapidly.

We also find evidence of a credit supply shock by looking at exporter firms. Several recent papers have documented the differential effect of financial crises on trade; exporter firms are more dependent on credit and so are more affected by a disruption to their banking partnership [Xu \(2021\)](#), [Manova \(2012\)](#), [Iacovone et al. \(2019\)](#). We find evidence that firms which produced tradable goods are driving the effect on bankruptcies.

We estimate equations (1) and (5) using only bankruptcies of firms producing tradable or

²¹According to [Crouzet \(1972\)](#) “short-term credit to finance increases in inventories was quantitatively by far the largest need of industry” during the industrial revolution.

Table 9: Estimation Results of Tradable and Nontradable Firm Bankruptcies

	(1)	(2)	(3)	(4)
	Tradables	Tradables	Nontradables	Nontradables
<i>BankExposure</i>	0.013*	0.096**	0.003	0.012
	(0.007)	(0.042)	(0.004)	(0.021)
Obs.	18500	18500	18500	18500
Districts	250	250	250	250
Mean	0.026	0.026	0.013	0.013
Model	OLS	2SLS	OLS	2SLS
Kleibergen-Paap		14.969		14.969

Notes: OLS and 2SLS estimation of equations (1) and (5). The dependent variable is the number of bankruptcies per 10,000 people of firms in sectors that sell tradable goods in columns (1) and (2) and nontradable goods in column (3) and (4). Errors are clustered by district in all columns.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

nontradable goods as part of the dependent variable.²² We report the results in Table 9. Our results match with the results in the literature; the estimates using tradable bankruptcies are positive and statistically significant, and the results using nontradable goods are insignificant and close to zero in magnitude.

6 Robustness

This section explores various robustness checks for the results already presented. We test various specifications that differ from our main model. We also use a placebo test to show that the instrument only predicts country bank failures during the crisis period. Lastly, we observe a positive correlation between districts connected to the financial network and patenting activity.

Changing the criteria that a district must have a bank in 1823 (our baseline rule) to a bank in 1825 does not change the results. Removing districts with multiple banks also does not substantially change the coefficient estimate though the instrument is now much weaker. This is likely because most of the observations are dropped - though the first stage is statistically

²²Estimating the share of tradable (or nontradable) firm bankruptcies is not feasible due to the sparsity of the bankruptcy data.

significant, the error term is much larger. Estimating the model using a continuous treatment variable, such that *BankExposure* measures the total number of bank failures in district i , rather than an indicator for the extensive margin, produces a statistically insignificant coefficient with roughly half the magnitude of the baseline estimation. While this could reflect that districts with multiple banks receive lower proportional weight from a single failed bank than in the discrete setting, recent criticisms of difference-in-difference estimation with a continuous treatment variable suggest that these results may not be credible [Callaway et al. \(2024\)](#).

All regressions based on the estimation of equation (1) or (5) cluster errors by district. Clustering errors by the larger geographic unit of county only marginally increases the error term, indicating that any serial correlation of the residuals is mostly at the district level. Similarly, using quarterly fixed effects instead of monthly does not affect the results. Removing outliers, defined as districts in the top 5% by bankruptcies per capita between January 1821 - February 1827, slightly reduces the coefficient but does not materially change the results.

To determine a bank failure, we observe if a bank has dropped out of the database between issues. This introduces a degree of measurement error as the window of time between issues during our period of interest is 1825-1827. We leverage our bankruptcy dataset to observe bank failures in December 1825 and January 1826 - the start of the Panic. Though this helps us measure bank failures more precisely across time, the bankruptcy dataset is not comprehensive, as discussed in Section 3. Estimating our main 2SLS model where we include only districts for which we observe a bank failure in these two months against a control group of districts which did not see a bank failure in 1825 or 1826 yields a positive coefficient for our estimated *BankExposure* variable though the F statistics is low and the estimate is insignificant. Given that we remove more than half of all observations (districts which we observe as having a bank failure between 1825-1827, but not in December 1825 or January 1826), this is unsurprising. Nonetheless, the positive coefficient is in line with our main results.

Aggregating the data to the quarterly level does not materially change the results. The coefficient of the baseline 2SLS result in column (1) of Table 5 increases to .331, reasonably

close to an expected three-fold increase. The reduced form and SDID estimates display similar results. As expected, the SDID event study plot is less noisy.

To further check the validity of the instrument and ensure that there were no systematic differences between London agents that did and did not survive the 1825 Panic, we use a placebo test. The probit models reported in Table 15 show that agent failure during the 1825 crisis did not predict earlier country bank failures between 1823 and 1825, after controlling for agent characteristics like the total number of clients the agent had and the agent’s total number of correspondent failures over the same period. Interestingly, in normal times (1823-1825), the failure of one of the bank’s local competitors reduced the probability it would fail, presumably because this expanded its business prospects. However, during the crisis the within-town contagion effect dominated and reversed the sign on “Bank failures in same district” (as seen in Table 3). This is a novel finding relative to the analysis of Calomiris et al. (2022) who find a consistently negative effect of the failure of other banks in town during the Great Depression in the U.S., perhaps because their analysis covers a protracted period with elevated bank run risk, whereas our sample covers both the boom and bust parts of the credit cycle.

In the previous section, we show that bank failures had a negative impact on the local economy. This shows the importance of finance to the British economy in the latter stages of the Industrial Revolution. However, it might be that country banks offered little value during good times and increased firm bankruptcies during bad times. In such a case, the financial sector would be seen as an unimportant (or even negative) aspect of the Industrial Revolution. We use historical patent data from Ravalli (2025) to examine the effects of the financial industry on the inventive side of the economy. Due to data constraints (namely, our country bank dataset ends in 1830 and the patent data are highly sparse prior to 1852), we restrict our analyses to correlations. To proxy the strength of a district’s links to the financial markets, we use a dummy variable equal to one if a district has a country bank in each year of data between 1820-1830 and zero otherwise.

We find that districts with stronger ties to the financial sector are associated with more patents per 10,000 people both during the 1823-1829²³ period concurrent with the timing of

²³The patent dataset begins in 1823.

our proxy variable and with the following ten year period.²⁴ Using the total number of banks in 1830 as a continuous variable also produces a positive, statistically significant coefficient. The coefficient on the log share of total failed banks between 1825-1827 is negative, implying that districts which experienced a stronger negative shock during the Panic had stronger lasting consequences between 1830-1839, though the coefficient is statistically insignificant.

7 Conclusion

This paper contributes to the literature on the role of finance during the Industrial Revolution. We find that financial integration in the form of the correspondent banking network played a critical role in transmitting financial stress induced by the Latin American debt crisis in 1825 to provincial economies. Our findings suggest that country banks played an important role in financing local firms: in the short run, the loss of a country bank caused an increase in local non-financial firm bankruptcies. Bankless districts seem to have been largely immune to the crisis suggesting that contagion spread through the financial network. These findings are consistent with models of failures in networks of interdependent financial organizations (see [Elliott et al. \(2014\)](#), for example). Such models predict non-monotonic effects of financial integration: integration initially allows contagion to travel farther, but eventually reduces individual organizations' exposure to their own idiosyncratic shocks. For example Scotland, with its more mature and well-integrated banking system, experienced much milder effects of the Panic of 1825 compared to England.²⁵ Geographic integration via correspondent banking may well have had positive effects in normal times; indeed, we find that districts connected to the financial network are associated with more patents during this period. Policy reforms in England in response to the Panic allowed banks to grow larger and expanded branch banking significantly, eventually enabling banks to more effectively smooth idiosyncratic local shocks.

Our findings also update arguments in the literature on the importance of finance in

²⁴We cut off the period window in 1839 to avoid the first wave of railroad construction which could affect these results.

²⁵[Calomiris and Haber \(2014\)](#) provide a comparison of the two countries' banking systems during this period. Scotland allowed joint-stock banking and Scottish banks were much larger than English country banks.

the industrial revolution in England. The rapid spread of bank failures to bankruptcies of non-financial firms suggests that banking services, particularly the means of payment they provided to households and firms, were important for the normal functioning of local economies at a short-term frequency. This point may be useful for understanding the potential consequences of rapid demonetizations like India's in 2016 as well as the potential costs of disruptions to cryptocurrency payment systems as these currencies become more widely used.

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A Data Appendix

A.1 Banking Network

As described in Section 3, the banking network data comes from the Post Office London Directories for five years: 1820, 1823, 1825, 1827, 1830. Historical records suggest these directories were published annually, but we were only able to locate and access these five years. The directories were published under the patronage of the postmaster general, and were meant to provide the names and addresses of government officials, merchants, and public companies. The relevant section in each year is called “A List of Country Bankers”, which contains the name (almost always the partner names), town, and London agent of all country bank branches that drew on a London bank.

We transcribed these records by hand since the number of entries was manageable, around 600 country bank branches per year. After transcribing each year separately we matched bank branches over time using their name and location. Country banks were limited to six partners, and partnerships changed somewhat frequently over time according to Pressnell (1956), so we chose to consider two branch-year pairs a match as long as the location and at least one partner name was the same. Within each year we also tried to match country banks branches across different locations to identify branches of the same banks. Here we used partner names and London agent, assuming that different branches would have the

same London agent. Around 2% of country bank branches had two London agents rather than one and we recorded each of these connections separately.

A.1.1 Identifying Agent and Bank Failures

As discussed in Section 3, our main identification of branch failures is based on branches that disappeared from the Post-Office Directories between 1825 and 1827 rather than on direct evidence. London agent failures identified using the same method match narrative evidence from [James \(2012\)](#) and [Dawson \(1990\)](#) closely (both sources list examples of London agents that failed during the crisis). We further use [Price \(1890\)](#)'s Handbook of London Bankers, which attempts to provide a comprehensive guide to all London banks in the 19th century, to verify when agents failed. In any case, agent failures are less likely to be mismeasured by this strategy since, unlike country bank branches which only appear once per year, each agent shows up in the Directories as many times each year as they have clients, which is nine on average.

A.2 Firm Bankruptcy Data

The bankruptcy data comes mainly from the Edinburgh Gazette. There are gaps in the Edinburgh Gazette publications available online, and we use the London Gazette for those cases. We collect data from January 1, 1820 - December 31, 1830. The start period for most analyses is January 1, 1821 to ensure maximum coverage with population data (as we are missing data on many districts' reported populations in the 1801 and 1811 Censuses; we also remove districts within Gloucestershire and Staffordshire due to missing population data in 1821 and 1831). The main crisis period is December 1, 1825 - February 28, 1827 and so we cut off most analyses on February 28, 1827. Excluding bankruptcies in London, we are left with 3,797 individual bankruptcies over the study period.

When an individual declared bankruptcy or was sued for bankruptcy by a creditor, a notice was required to be posted in the London, Edinburgh, and Dublin Gazettes so other creditors were aware of the proceedings. [Marriner \(1980\)](#), an authority on bankruptcy statistics from this period, argues that Gazette records provide an accurate picture of bankruptcy

statistics because the government-appointed Commission of Bankrupt had to certify that creditors had a legitimate claim before notices were posted in the Gazettes. [Duffy \(1973\)](#) analyzed 50 bankruptcies from 1810-1811 and found that for 41 out of 50 cases of payment stoppage, bankruptcy proceedings began within one month, so the dating is fairly accurate.

By searching all banker names in the Post-Office Directories that we identify in those databases as bankrupt, we find that some bankers' occupations are not listed as banker in their bankruptcy notices in the Gazette. This misclassification could introduce bias by understating banker failures and counting them as non-financial firms, so we drop bankruptcies of individuals with the same name as the bankers who failed during the Panic, but we may miss some since not all partner names are listed in the London Directories. For banks with two or more partners listed, however, we actually find few cases where more than one partner went bankrupt.

A.2.1 Occupation Classifications

We classify the occupations of bankrupt partners into tradables, nontradables, construction and other. We generally only use the first listed partner's occupation for the classification. General dealers, brokers, and merchants are classified as tradables. Nontradables tend to be occupations dealing in finished food products, such as bakers, and services occupations.

A.2.2 Focus on England

As discussed in Section 3, we focus exclusively on English bankruptcies rather than Irish or Scottish bankruptcies. This is acceptable because both countries differed from England in important ways. In Scotland, for example, joint-stock banks were not prohibited, banks had limited liability, and most banks had many branches,²⁶ more closely resembling modern day financial systems and allowing Scottish banks to better weather crises like the 1825 panic.²⁷ Both Scotland and Ireland had their own (proto-)central banks, adding an additional layer to the institutional differences between these national contexts. It is not clear how and

²⁶[Black \(1995\)](#) provides a useful description of how the Scottish system differed from the English system.

²⁷The *Edinburgh Courant*, quoted in the December 25th, 1825 edition of [The Examiner \(1825a\)](#), wrote "The consideration of these circumstances forces upon our notice the superior security which our Scottish banking establishments afford...the alarm of the money-market in London has scarcely been at all felt."

where Welsh bankruptcies were reported, but we find very few Welsh bankruptcies reported in the Edinburgh Gazette despite Wales having a large number of country banks, so we also exclude Wales.

A.2.3 Geographic Data

We use the OpenCageGeocoder API to find the coordinates for town names listed in both the banking network and the bankruptcy datasets. We then manually verified each of these using Google Maps. The coordinates were then matched with a shapefile of English district boundaries provided by [Satchell et al. \(2023\)](#) to pinpoint the towns' district. We use district level data as the population across time is much more accurate than at the town level.

A.3 Additional Tables

Table 10: Bankruptcies in England from Edinburgh Gazette by Occupation Category

	1825	1826	1827
Tradables	186	654	180
Non Tradables	109	268	98
Construction	35	95	36
Other	60	105	64

Post-Office London Directories, 1820-1830.

Table 11: Comparison of Exposed vs. Not Exposed Banks, 1825

	Exposed Mean	Not Exposed Mean	Difference Diff.	t-stat
Number of bank branches	2.56	1.73	0.83***	2.95
Has more than one agent	0.02	0.03	-0.01	-0.68
Founded 1821-1825	0.27	0.26	0.00	0.08
Number of banks in district	2.12	2.31	-0.19	-1.47
District population, 1821, in thousands	21.52	26.13	-4.61**	-2.39
Firm bankruptcies in district, pre-period	1.98	2.39	-0.40	-0.63
Observations	120	418	538	

Source: Post-Office London Directories, 1820-1830; Edinburgh Gazette; [Mitchell and Jones \(1971\)](#). Pre-period for firm bankruptcies is January 1, 1825 - November 30, 1825. Firm bankruptcies exclude banks. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Robustness Exercises - Banks

	(1)	(2)	(3)	(4)	(5)
<i>BankExposure</i>	0.138** (0.061)	0.134** (0.058)	0.153* (0.092)	0.044 (0.030)	0.016 (0.013)
<i>BankExposure</i> X Old Banks					0.014 (0.015)
Obs.	18500	18574	7992	18500	18500
Districts	250	251	108	250	250
Mean	0.058	0.057	0.037	0.058	0.058
Analysis	Baseline	Has Bank, 1825	Only One Bank	Cont.	OLS
Kleibergen-Paap F	14.969	16.214	4.197	8.501	

Notes: 2SLS estimation of equation (5) which includes district and monthly fixed effects. The dependent variable is the number of non-financial firm bankruptcies in district i and month t . Column (1) reports the baseline results from Table 5. Column (2) excludes districts without at least one bank in 1825 (rather than 1823 as in the baseline). Column (3) only includes districts with a single bank. Column (4) changes the definition of $BankExposure_{i,t}$ such that it is calculated using the total number of banks (and agents for the instrument) in district i , rather than measuring the extensive margin. Column (5) estimates the OLS reduced form model and includes an interaction between $BankExposure_{i,t}$ and an indicator equal to one if a district has at least one bank that operated between 1820-1825 and zero otherwise. Errors are clustered by district in all columns. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Robustness Exercises - Econometric

	(1)	(2)	(3)	(4)	(5)
<i>BankExposure</i>	0.138** (0.061)	0.138** (0.062)	0.101* (0.054)	0.124** (0.048)	0.161 (0.124)
Obs.	18500	18500	17612	18500	8806
Clusters	250	38	238	250	119
Mean	0.058	0.058	0.050	0.058	0.053
Analysis	Baseline	Cluster County	Outliers	Quarterly FE	Precise Fail
Kleibergen-Paap F	14.969	10.692	13.230	23.855	3.034
Time FE	Month	Month	Month	Quarter	Month

Notes: 2SLS estimation of equation (5) which includes district and monthly fixed effects. The dependent variable is the number of non-financial firm bankruptcies in district i and month t . Column (1) reports the baseline results from Table 5. Column (2) clusters errors at the county level rather than district as in all other columns. Column (3) excludes districts in the top 5% by the total (across Jan 1821 - Feb 1827) bankruptcies per capita. Columns (4) includes quarterly fixed effects instead of monthly. Column (5) includes only districts which did not suffer a bank failure between 1825-1827 and districts for which we observe a bank failure in the bankruptcy data for the months December 1825 and January 1826. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Robustness Exercises - Quarterly

	(1)	(2)	(3)	(4)
<i>BankExposure</i>	0.331** (0.152)	0.512** (0.239)		
<i>AgentExposure</i>			0.075** (0.031)	0.061* (0.033)
Obs.	6250	5500	6250	6250
Districts	250	250	250	255
Mean	0.173	0.167	0.173	.059
Analysis	SLS	SLS	Reduced Form	SDID
Last Quarter	1827 Q1	1826 Q2	1827 Q1	1827 Q1
Kleibergen-Paap	14.971	14.971		

Notes: This table replicates the results from Table 5 but the data are aggregated at the quarterly level instead of the monthly level. Quarterly fixed effects are included in every column and errors are clustered by district. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 15: Placebo Probit Models for Bank Failure, 1823-1825

	(1)	(2)	(3)	(4)	(5)
Bankruptcy, 1823-1825					
Agent bankruptcy, 1825	0.130 (0.174)	0.136 (0.179)	0.140 (0.179)	0.153 (0.183)	0.073 (0.200)
Number of other branches		0.001 (0.040)	-0.001 (0.040)	0.017 (0.038)	0.018 (0.050)
Founded 1821-1823		0.597*** (0.168)	0.597*** (0.168)	0.633*** (0.171)	0.708*** (0.187)
CB bankruptcies in same city, 1823-1825			-0.119 (0.192)	-0.124 (0.198)	-0.516*** (0.194)
Agent's number of clients, 1823				-0.002 (0.007)	0.003 (0.008)
CB bankruptcies of same agent, 1823-1825				-0.046 (0.053)	-0.063 (0.058)
County FE					Yes
Observations	569	569	569	569	396

Note: Data from the Post-Office London Directories, 1820–1830. Country bank branch level probit model where the dependent variable is equal to 1 if country bank branch i fails between 1823–1825. The number of observations differs from Table 3 due to differences in bank numbers across time. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Correlation between Banking and Patents

	(1)	(2)	(3)	(4)
Has Bank, 1820-1830	0.155** (0.076)	0.294*** (0.111)		
Total Banks, 1830			0.169** (0.066)	
Log Sh. Failed Banks, 1825-1827				-0.387 (0.378)
Obs.	507	507	507	174
F Stat.	4.166	7.033	6.496	1.051
Mean Dep. Var.	0.312	0.548	0.548	0.926
Dep. Var. Years	1823-1829	1830-1839	1830-1839	1830-1839

Notes: OLS regression between the number of patents per 10,000 people as the dependent variable and the bank status of the district as the independent variable. In column (1) we measure the aggregate number of patents between 1823-1829 and the 1821 district level population and in columns (2)-(4) the number of patents between 1830-1839 and the 1831 population. The independent variable in columns (1) and (2) is an indicator equal to one if district i has at least one bank in every year between 1820-1830, and zero otherwise. In column (3) it is the total number of banks in a district in 1830 and in column (4) it is the log share of failed banks between 1825-1827. Only districts with at least one failed bank between 1825-1827 are included in column (4). Robust errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A.4 Additional Figures

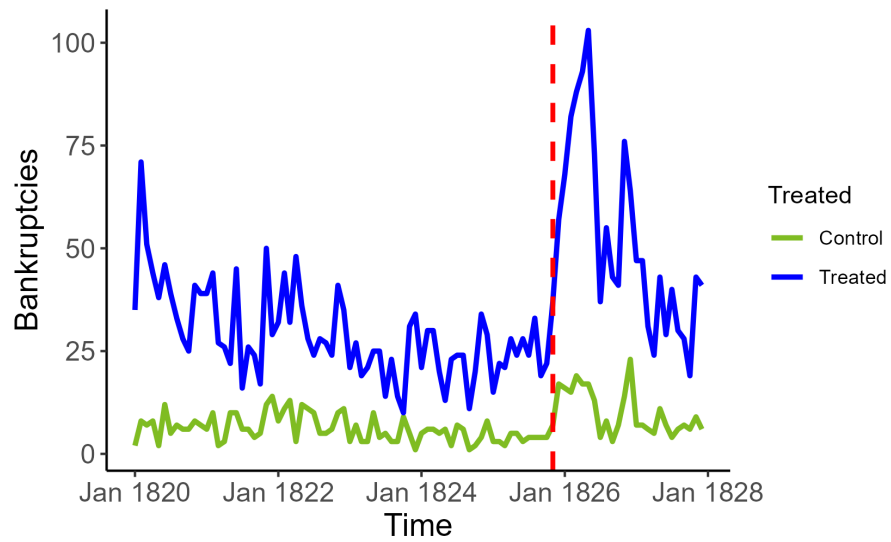


Figure 7: Number of non-financial firm bankruptcies across time where “Treated” bankruptcies occur in districts with a bank failure between 1825-1827 and “Control” bankruptcies occur in districts with a bank in 1823 but without a bank failure. The red dotted line marks November 1825, the month before the start of the Panic.

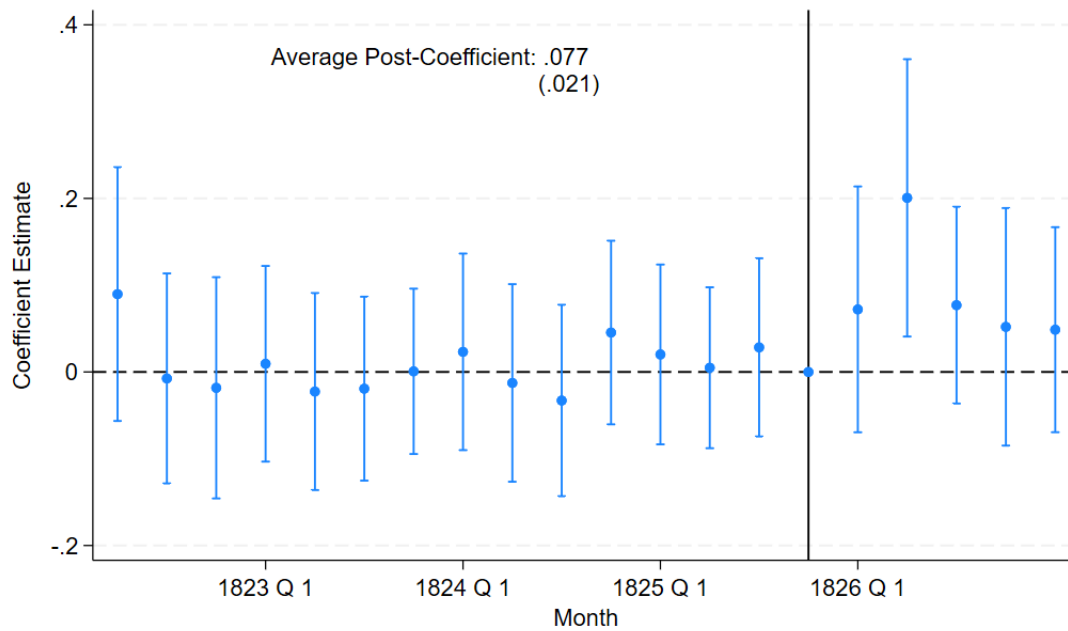


Figure 8: Quarterly version of Figure 4.

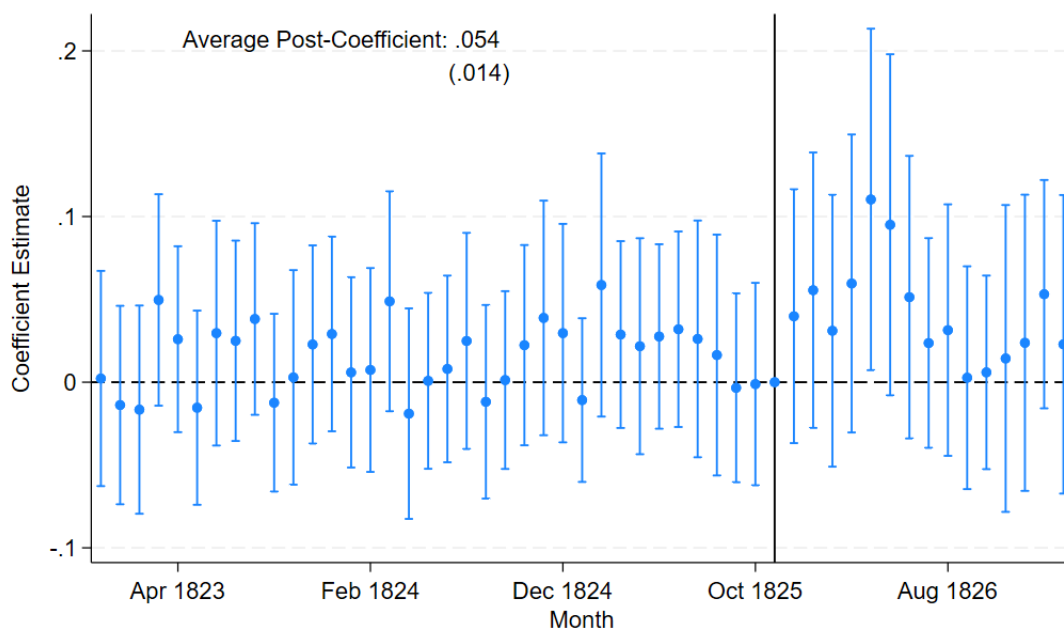


Figure 9: Reduced form version of Figure 4 using London agent failures as the instrument.

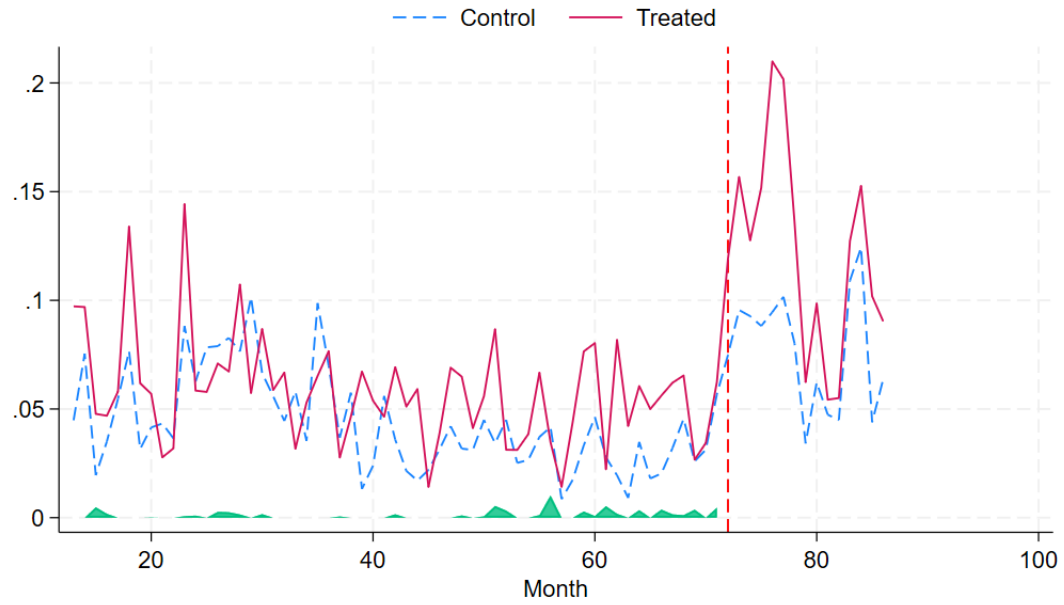


Figure 10: Synthetic difference-in-difference event study plot at the monthly level of the reduced form estimation of equation 5.

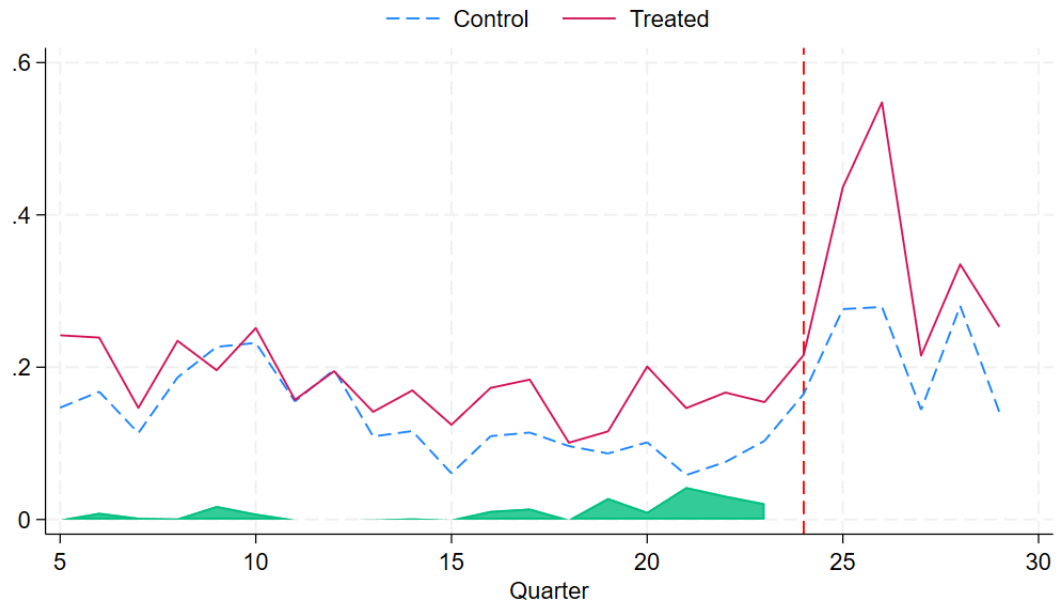


Figure 11: Quarterly level version of Figure 10.

A.5 Maps

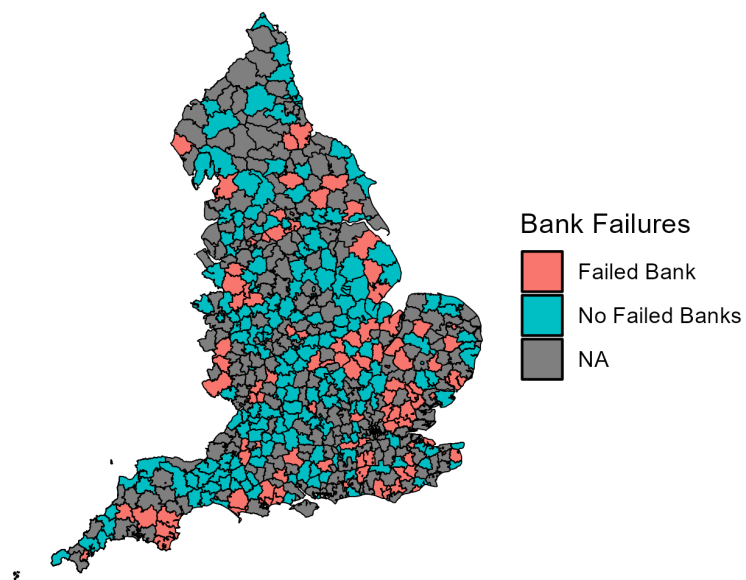


Figure 12: District level bank failure status for failures that occurred between 1825-1827. NA indicates that there were no failed banks in the district.

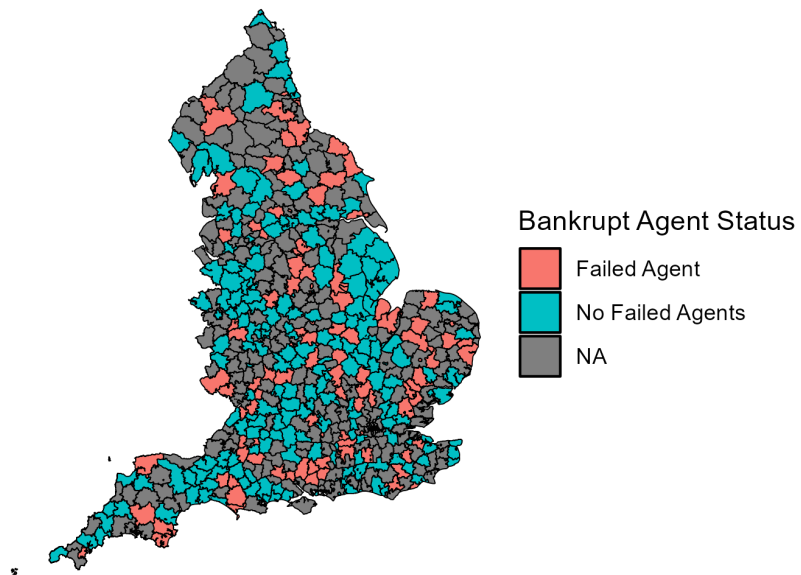


Figure 13: District level agent-driven bank failure status for failures that occurred between 1825-1827. NA indicates that there were no agent driven bank failure in the district.

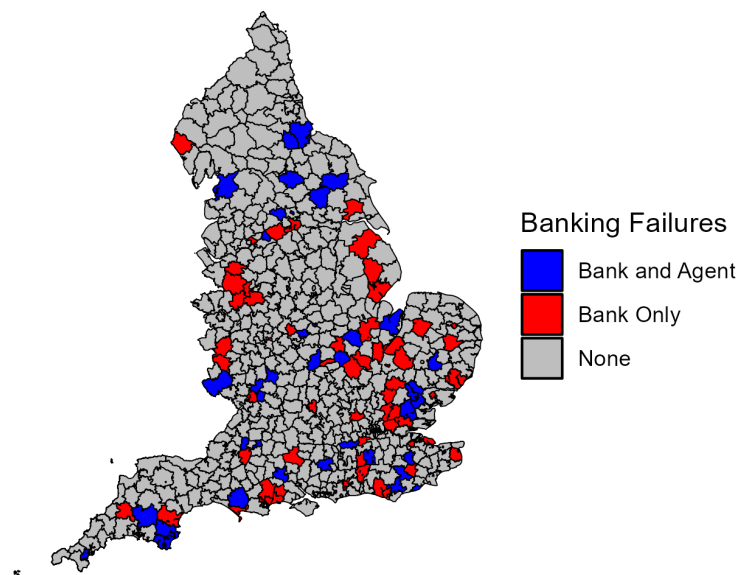


Figure 14: District level bank failure status. Blue indicates a district had at least one bank with a failed agent and at least one failed bank during the Panic; red indicates at least one failed bank and no banks with a failed agent; gray indicates no failed banks or agents.

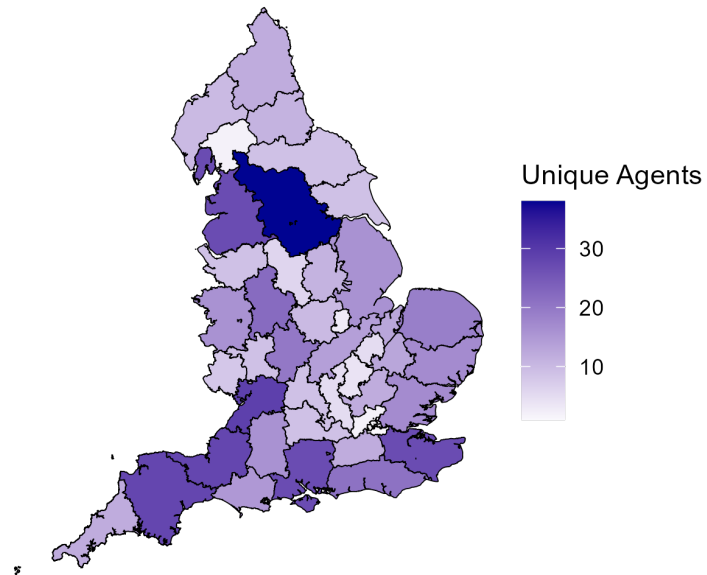


Figure 15: The number of unique agents with a local banking relationship in a county.

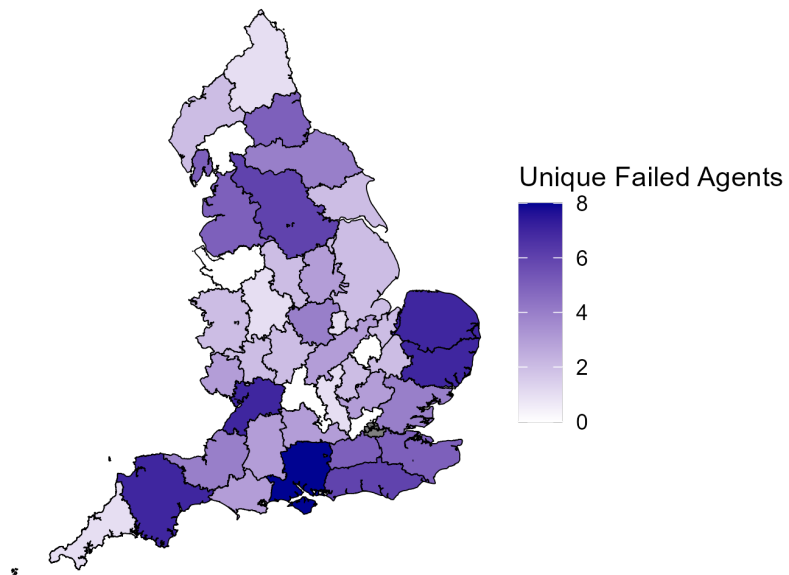


Figure 16: The number of unique agents that failed during the Panic with a local banking relationship by county.

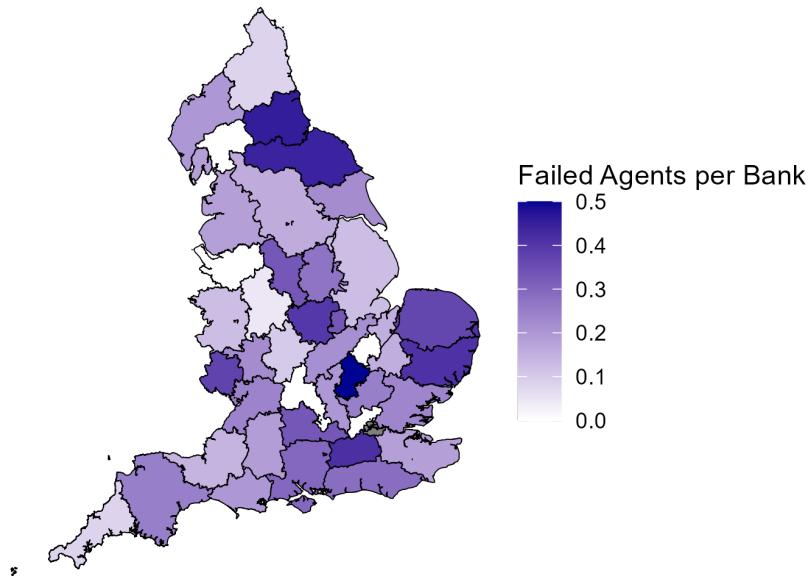


Figure 17: The number of unique failed agents per country bank in each county.